2019

CALENDAR YEAR REPORT

WASTEWATER TREATMENT FACILITIES

City of Memphis

Division of Public Works

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Wastewater Treatment Facilities
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1.0 Maxson Wastewater Treatment Facility

Introduction- The Maxson Facility was met with an avalanche of activity this year. The kickoff meeting for the replacement of the primary sludge 18-inch force main was on 1-3-2019. The original ductile iron pipe had lasted 14 years before starting to fail with wear evident in the form of a narrow cut along the invert of straight sections of the pipe. On 1-22-2019, a slug discharge created a drop in mixed-liquor dissolved oxygen late in the afternoon and elevated effluent BOD and TSS and lower SVI's over the next several days. On 1-23-2019, a raw sewage pump failed due to overheating related to the IPA building which apparently was not meant to house VFDs. A meeting was held with USP Technologies on 1-24-2019 to discuss a 90 day demonstration project at Lagoon No. 5. The project would evaluate the use of peroxide to oxidize sulfides and other odor constituents including ammonia.

Slug discharges with strong petroleum smell were received 2-4-2019 through 2-8-2019 and 2-11-2019 through 2-15-2020, causing foamier effluent than normal. CDM Smith had to stop re-bar work on the GMP 1 disinfection tank on 2-21-2019 due to high Mississippi River stages. The Mississippi River Stage crossed over 38 feet on 2-24-2020, beyond which the Maxson Facility average daily flow increases approximately 18 mgd per every foot of rise. TDEC held a public meeting on 2-26-2019 for the Maxson draft NPDES permit and granted an extension to receive public comments to 3-13-2019. The City requested that industrial categories without promulgated effluent limits be used for secondary treatment adjustment calculations based on BPJ. The loss of Cargill's fructose operation in 2015 which had promulgated limits based on production necessitated this approach. On 2-27-2019, a sinkhole was observed formed in the trailer storage yard east of the Maxson Facility, south of the Nonconnah Interceptor and appeared related to a 12-inch sewer outfall

A possible slug discharge observed on 3-6-2019 and remaining in the activated sludge for a few days resulted in elevated effluent BOD and TSS. The sinkhole started getting larger on 3-13-2019 as Nonconnah Interceptor became less surcharged with falling river stages. This caused an influx of sand resulting in a bearing failure on Primary Clarifier #3. A new cavity then formed at manhole #7 at the Nonconnah Interceptor just east of Plant Drive. This was related to the 24-inch lagoon and dewatering filtrate line installed in about 1996 when secondary dewatering was constructed. Environmental Maintenance continued adding sand to the sink hole in the trailer yard in the hope that it would eventually bridge over. The majority of the sand began either entering the Maxson Facility grit tanks and primary clarifiers or becoming lodged upstream of the raw sewage pump channels or being pushed upstream and settling in the Horn Lake Interceptor. While trying to repair the 24-inch filtrate line, Environmental Maintenance broke the 18-inch primary sludge wasting line which prevented sludge wasting for a few days. It was discovered that the manhole junction inlet was sized for a 36-inch instead of a 24-inch pipe. It had been rigged to receive a 24-inch pipe and eventually came loose. Once the 18-inch wasting line was repaired, it was observed that its discharge was doubling-back into the activated sludge wasting line if the activated sludge wasting pump was not operating. On 3-22-2019, the Mississippi River stage receded below 38 feet and the Maxson Facility ADF fell to normal dry

weather flow. The plant suffered a lightning strike on 4-7-2019 taking out a card to RSP#2 which required the attendance of a factory technician to repair.
On 4-22-2019, RSP#3 went down on vibration. The 12-inch outfall line in the trailer yard was taken out of service by Memorial Day and the TVA Combined Cycle sewer line (non-process) was permanently connected to the Maxson Facility 24-inch filtrate line.

The final walk-through for the Secondary Dewatering Building skin replacement was on 6-26-2019. On 6-26-2019, both MLGW substations #82 and #41 (backup) shut down during a lightning storm. MLGW restored power by back feeding from the Weaver Road substation just before the plant bypass was to be operated.

A contract with Tri State was executed on 7-5-2019 in the amount of $225,075 to complete the upgrade on the remaining four blowers with expected completion by 12-31-2020. A partial bypass was implemented on 7-10-2019 in conjunctions with the startup of the headworks bypass for the GMP 2 project. The Horn Lake Interceptor gate was closed on 7-22-2019 and the full bypass started on 8-6-2019. This caused an upstream disturbance and resulted in an avalanche of grit entering the plant.

The Maxson Facility received a Notice of Violation on 8-9-2019 for Significant Non-Compliance from December 2018 through June 2019. TDEC has been criticized in the past by EPA for not flagging DMRs and issuing NOVs using SNC review criteria. Approximately 8 dump truck loads of sand and debris were removed from the influent channels after startup of the headworks bypass.

The first set of coarse bar screens arrived on 9-26-2019. An inspection of the wet well showed corrosion worse than expected including severe corrosion related to the return sludge discharge into the wet well between RSP channels #1 and #2. Original wall coating had been installed in 2000. Two 12-inch and 24-inch plugs were found in the RSP#4 pump volute which had been out of service since March 2018 due to excessive noise. Obviously, these plugs were previously lost from an upstream sewer construction project in either the Nonconnah Creek or Horn Lake Basins.

On 10-31-2019, the new Mud Cat dredge arrived and eventually put into service in Lagoon No. 5. On that same day, the Maxson Facility suffered a slug discharge consisting of a milky white substance smelling like solvent and French fries which caused an increase in mixed-liquor dissolved oxygen concentration and elevated effluent BOD and TSS. Discharges of a less intense nature had been received at weekly intervals throughout the month. Similar discharges in the past have been associated with Memphis Bioenergy.

Demolition of necessary interior of the screen building was complete and work continued on concrete repair in November. Additional settlement of Secondary Clarifier 3N is suspected as a result of the sustained high river stages for the first half of the year. The last settlement occurred in 2011 and was subsequently stabilized as a FEMA project.

The Stiles Facility lab began was relocated to the Maxson Facility on 12-3-2019 after refurbishment of the former Maxson Facility lab was complete. The idea is for them to remain there while the Stiles Facility lab is renovated during the upcoming year (2020). The GMP 2 head works pump-around was extended through 2-6-2020 ostensibly to deal with the unexpected corrosion repair although the additional time would also be utilized to perform original scope work. The Notice to Proceed for GMP 3 was issued on 12-19-2019 for Package 2A. The year ended with a major pipe gallery leak on 12-27-2019 which was temporarily stabilized and subsequently repaired by a contractor specializing in pipe wrapping in January. Two blowers malfunctioned in December due to oil leaks leaving the Maxson Facility with three functioning blowers. The repairs were complicated because the blower motors required unique adaptor plates. As a result, there is growing interest in replacing the oil lubricated with grease lubricated blowers due to reduced maintenance. Maxson Facility maintenance and operations personnel continued manning the effluent pump station on a 24/7 basis throughout the year to prevent any mishaps that would back effluent into the disinfection tank while under construction.

The Maintenance Department performed numerous noteworthy tasks during the year. In January, at the East Lift Station, staff installed two new 12-inch suction side gate valves and two new 8-inch and 12-inch check valves, retro-fitted and drilled out both volutes motor base bolt holes and inserted 7/8" Heli coil. Both north and south motors and impellers were refurbished and ready for installation. At the Horn Lake Metering Station, staff fabricated and installed stainless steel access ladder, hand railing, safety tie off points, and access panel, removed brackets, aluminum floor and channel grating and installed new bubbler system for channel level reading. In February, at the Admin Building, crews primed and painted the entry gate, iron works and signage at the front of the building. At the Pipe Gallery staff replaced 60 feet of 12-inch effluent ductile iron pipe inside pipe gallery and replaced and relocated 120 feet of the same pipe outside of Pipe Gallery (south side). In the Maintenance Shop Building, crews cleaned and painted walls and floor of maintenance shop. At the Lagoons, crews rebuilt all blowers on Lagoon 1A. Staff replaced suction bell sleeve, sandblasted, primed and painted entire pump housing, using newly fabricated shafts, couplers and impeller, and assembled pump at #4 ABF Pump. At #3 South Blower Control, crews completed blower control renovations. A new motor was purchased and installed on #3 South Blower. In March, at #3 Fine Bar Screen, maintenance staff replaced motor, broken AM adaptor and installed new pin rack rollers. At the Pipe Gallery Effluent bypass line, they installed approximately 175 feet of 12-inch ductile iron pipe and fittings inside/outside pipe gallery to move pipe from being underneath newly poured concrete foundation. A new 4'x4'x6' kicker bracing to secure pipe at elbow was fabricated and installed. At the Lagoon, crews completed submersible pump rebuild, epoxy coated tank, and installed new railing and grating for Submersible Pump. Backflow Preventers and failed systems were repaired and certified throughout plant and lagoon. At #2 ABF Motor, crews rebuilt failed motor drive, and replaced upper and lower motor bearings.

In April, at the Maintenance Shop Building, old vinyl floor tile was removed from breakroom and hallway and 1,050 square feet of porcelain tile was installed, including grout and new base board. At #2 Screw Pump, crews rebuilt the gearbox, repaired the output shaft and changed out seals. Crews installed new undercarriage and drive seals on PW304 Cat Dozer. Crews replaced upper and lower motor bearings and seals on #7 ABF motor. Crews overhauled #5 Dewatering Press, including replacing rollers, belts and motors. Crews installed 12-inch drain box at 110 South at the Lagoon. At #1 South Blower Motor, a complete motor rewind was performed and new bearings installed. In May, crews removed, completely rebuilt and reinstalled pump at #4 ABF Pump and put back into service. At #7 ABF, crews rebuilt failed motor drive and replaced upper and lower motor bearings. The stator lamination was replaced and stator rewound and reinstalled on #4 Pit Hog. The electrical pump on #2 Pit Hog was removed and refurbished. In June, a complete re-skin of exterior of Secondary Dewatering building was performed using preengineered metal. At #1 Screw Pump, the gearbox out-put shaft was machined, all new bearings and seals were installed, and a new backstop was purchased and installed (complete rebuild). At #2 Screw Pump, the flight assembly was removed because the bottom end cap on the flight assembly had started to break away, then repaired and reinstalled.

In July, at #8 ABF, the failed motor drive was rebuilt and upper and lower motor bearings were replaced. A new engine and new undercarriage were installed on PW309 Cat Dozer. In August, the gate into the Horn Lake Interceptor was refurbished. 125 feet of 42-inch pipe was exposed for future replacement behind Lagoon 2B. In September, two new clarifier center drives, one primary and one secondary drive, were procured.

In October, a new 10-ton HVAC unit was installed on the 23 KV Building. A new motor for #1 North Blower was purchased and installed and the oil and water piping on blower motor platform was reconfigured to accommodate new style motor. At #3 Primary Clarifier, the clarifier arms were refurbished, a new center drive installed, and new arms were fabricated and installed for center drive connection brackets. At the Old Chlorine Building, the concrete drive and storage area were re-concreted. 103 feet of fence was erected around concrete pad and door opening enlarged to accommodate fork lift traffic. A new motor for #2 North Blower was purchased and installed and the oil and water piping on blower motor platform was reconfigured to accommodate new style motor.

In addition to the major tasks listed above, the Maintenance Department completed 3514 Preventative Maintenance Work Orders, 871 Scheduled Maintenance Work Orders, and 1756 Unscheduled Maintenance Work Orders for a total of 6141 work orders (including some not categorized). The annual number of work orders completed over the last 10 years is summarized as follows:

Year	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
Preventative	3514	3417	1896	178	567	1000	1236	1777	1285	1661
Scheduled	871	799	321	20	72	96	165	263	185	109
Unscheduled	1756	1190	252	18	118	125	138	596	658	739
Total	6141	5416	2469	265	772	1234	1784	2650	2128	2509

Ferrous Chloride Use –No ferrous chloride has been used at the Maxson Facility since January 2012. Among its varied uses, ferrous chloride reduces the likelihood of struvite formation by producing acidic conditions and binding up magnesium and phosphorus ions. Struvite formation began to be observed in Lagoon Nos. 3 and 4 in 2015, favoring areas where there is turbulence. A 5000 gallon tank has been placed into position to add ferrous chloride to feed sludge prior to dewatering when this practice is resumed in this more limited manner. No ferrous chloride was purchased or used in 2019. In the future, purchase options would include commercially made ferrous chloride containing about 34% or spent pickle liquor containing 10% iron.

Flow – The average daily flow rate in 2019 was 78 mgd which was the same as the previous year. The 0.5th percentile flow in 2019 was 49 mgd which is considered base flow for the year. The average daily flow received from significant industrial users was 9.6 mgd. This leaves 19.4 mgd which is 25% of average daily flow unaccounted for most of which is related to river and rainfall (including run-off from uncovered lagoons and the fields). The total rainfall for the year was 74 inches (NWS). Memphis annual rainfall has averaged 53 inches.

BOD_5 – The average influent BOD_5 loading for 2019 was 502,760 pounds per day, which is 32,240 pounds or nearly 6.4% decrease from the previous year. The average influent BOD_5 concentration was 788 mg/l and the average effluent concentration was 40 mg/l, resulting in a removal efficiency of 95%. The total average BOD_5 reported being discharged by the major significant industrial users was 272,845 pounds per day in 2019, equivalent to 54% of the total loading. This is 7540 pounds or 3% more in monitored industrial loading from the previous year. The 0.5^{th} percentile loading received in 2019 was 155,828 ppd which is the assumed domestic loading for this year. This leaves at least 74,087 ppd of influent BOD_5 which is 15% of average influent BOD loading unaccounted for.

Total Suspended Solids (TSS) – The influent TSS averaged 412,950 pounds per day, which is 113,700 pounds or a nearly 27.5% decrease from the previous year. The average influent TSS concentration was 646 mg/l and the average effluent concentration was 42 mg/l resulting in a removal efficiency of 93%. The total average TSS reported discharged by the major significant industrial users was 151,398 pounds per day in 2019, equivalent to 37% of the total loading. This is 19,918 pounds or 13% decrease in monitored industrial loading from the previous year. The 0.5^{th} percentile loading for 2019 was 102,869 ppd which is expected domestic loading for this year. Lagoon supernatant and dewatering filtrate return was calculated to be 14,953 ppd. This represented 3.6% of influent TSS received in 2019. This leaves 143,730 ppd of influent TSS which is nearly 35% of average influent TSS loading unaccounted for.

Chemical Oxygen Demand (COD) - The influent averaged 1514 mg/l and 965,859 ppd. The effluent averaged 184 mg/l and 120,513 ppd. The average removal was 88%. The 0.5^{th} percentile loading for 2019 was 261,681 ppd which is expected domestic loading for this year. The total average COD discharged by the major significant industrial users was 544,167 ppd which equals 56% of the total loading. The remaining 160,011 ppd or 17% of influent COD is unaccounted for.

Effluent E. coli - E. coli analysis was performed on daily effluent grab samples throughout the year as required by the NPDES permit as report only. Effluent E. coli results for this past year are presented in Figure 1 with apparent seasonal trends.

Water – In 2019 metered water usage was 60,025 CCF per month resulting in total usage of 720,294 CCF or 1.5 MGD. The total annual cost was $720,294. Actual annual potable water consumption is estimated at 823,812 CCF or 1.7 MGD due to the water meter at secondary dewatering being out of service for the first five months of the year. This is a decrease of 16,734 CCF per month from the previous year likely due to decreased secondary dewatering production.

Energy – The main power consumption of the plant was an average of 5,900,050 kwh per month in 2019. This is 247,070 kwh more per month than the previous year. The monthly peak demand averaged 9091 kW which is 363 kW more than the previous year. The average monthly main power bill was $424,827, which is $24,810 more than the previous year. Payments for participating in the ENERNOC curtailment program for main power totaled $40,496. March was the high month for demand at 10,027 kw and for consumption at 6,190.366. December was the low month for demand at 8275 kw and for consumption at 5,228,366 kwh. Hourly power consumption throughout 2019 is shown in Figure 2. The months September through December had estimated values due to unavailability of power meter as a part of GMP 1 electrical upgrades

Effective 10-1-2015, the City entered into an agreement with MLGW to switch from Seasonal Demand to Time of Use billing. TVA would charge Fuel Cost Adjustment based on user class. TOU rate structure benefits high load factor customers through incentivizing off-peak energy usage. The contract term is 5 years and will be automatically extended unless otherwise terminated.

Sludge and Biosolids – The quantity of primary sludge wasted to the lagoons in 2019 was 91.7 million pounds, a decrease of 21.8 million pounds from the previous year. The quantity of secondary sludge wasted to the lagoons in 2019 was 30.4 million pounds, an increase of 5.6 million pounds from the previous year. The total quantity of both primary and secondary sludge wasted from the plant was 122.7 million pounds, a decrease of 16.2 million pounds from the previous year.

Approximately 15.8 million pounds of digested sludge was dewatered at the secondary dewatering facility and disposed in the 800 acres available for Surface Disposal, of which 309 acres is considered to be permanently disposed. This includes an unknown amount of ferrous chloride residual from previous years that was added to Lagoon Nos. 3, 4, and 5 (the uncovered lagoons). Approximately 50.5 % of the combined primary and secondary sludge wasted to the lagoons in 2019 was destroyed equivalent to roughly 62 million pounds. The formula used to calculate percent destroyed is %VS of wasted primary and activated sludge multiplied by %VAR. The %VS for primary and waste activated sludge is based on two separate sampling points and then combined mathematically based on proportion. The %VS for feed sludge is based on a single sampling point.

Considering land filling and volatilization the new change in the lagoon solids inventory was an increase of 39.4 million dry pounds within the approximately 257 acres of covered and open lagoons. The calculated cumulative amount of dry solids within the covered and uncovered lagoons is 420.5 million dry pounds. The calculated amount of dry bio-solids that has been surface disposed since 1979 is 1410 million dry pounds.

A total of 77.9 million pounds of digested sludge was disposed in 2019 as surface disposed biosolids (7900 tons) and volatilized lagoon sludge (31,000 tons) at a total cost of approximately $4,340,267. This equates to a unit cost of $112 per dry ton. The annual cost per dry ton for surface disposal and volatilization combined for the past 10 years is as follows:

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
$112	$67	$66	$76	$68	$68	$62	$68	$59	$49

Generally, the unit cost to dewater and dispose sludge as bio-solids is a function of polymer cost and dewatering throughput. The cost to dewater digested sludge and surface dispose as bio-solids (i.e. dewatered digested sludge) in 2019 is estimated to be $549 per dry ton. The annual cost to dewater digested sludge and surface dispose as bio-solids per dry ton for the past 10 years is as follows:

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
$549	$265	$222	$201	$214	$262	$190	$247	$145	$100

The annual amount of digested sludge (in million dry pounds) removed from the lagoons over the last 10 years is provided below:

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
15.8	23	24.7	29.95	27.2	23.4	31.2	24.1	41.67	56.3

For the last 10 years, net accumulation has been occurring at an average rate of 42.6 million dry pounds per year. Surface disposal has averaged 29.7 million dry pounds per year. In order to stop net accumulation, the Maxson Facility will need to dewater and surface dispose an average of 72.3 million dry pounds per year which is nearly 2 ½ times the existing rate and would be 97 percentile level of performance.

Biogas – For the year 2019, the Maxson Facility produced a total of 552 million cubic feet of biogas. TVA used zero million cubic feet of the annual production.

The annual biogas produced by the Maxson Facility for the past 10 years is as follows (in millions of cubic feet):

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
552	565	744	1167	1254	1155	1027	1085	1015	851

The annual daily average biogas produced for the past 10 years is as follows (in millions of cubic feet):

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
1.51	1.55	2.04	3.20	3.44	3.16	2.81	2.97	2.78	2.33

Annual biogas yield for the last 10 years (ratio of biogas to volatiles destroyed) in cubic feet per pound is as follows:

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
8.9	8.3	12.4	23.8	21.4	14.1	12.5	13.2	11.2	14.7

The amount of sludge volatilized each year is relatively constant with no discernable long-term trends or impact on the volume of biogas produced. What is variable is the amount of biogas produced per pound of sludge destroyed as shown above. Since 2017, biogas volume has dropped to levels not seen since 2003 when Lagoons #2A, 2B, and 2C were first covered. Trends in biogas production over the years are shown in Figure 3.

A Construction Permit for the new flares was submitted to the SCHD on 11-23-2016. A revised permit application was submitted on 1-17-2017 to include the biogas flows and emissions to conform to CDM Smith's design numbers. The SCHD issued a construction permit on 4-10-2017, expiring on 10-11-2018. The City provided comments to the SCHD regarding the permit conditions on 5-5-2017. The City transmitted a Title V operating permit application to the SCHD on 5-18-2018 due to the Major Source designation based on potential air emissions of CO in excess of 100 tpy. Deficiencies were noted in the application submittal and subsequently corrected on 6-11-2018.

The City requested a construction permit extension on 1-15-2019 and it was granted through 12-31-2019. On 3-22-2019, the City requested a change in the VOC emission factor from Industrial Flare to Digester Gas Fired Flare. The City met with the SCHD on 4-1-2019 to discuss additional requirements that were missed in the construction permit application. SCHD made a follow-up request on 4-5-2019 for updated Title V SO2 emissions as an amendment to the original Title V application, as well as actual SO2 emissions for 2016-2018, and a Capacity Assurance Monitoring Plan for the Biorem system in accordance with 40 CFR Part 64. The City submitted the revised SO2 emissions on 4-30-2019 and the CAM Plan on 5-23-2019. SCHD public noticed a draft Title V air permit in September. The City provided final comments on 9-20-2019 as well as a revised CAM Plan. SCHD issued the final Title V permit on 10-4-2019, Permit # 00701-01TV, expiring 10-4-2024. A new permit application would be required by 4-27-2024. SCHD issued a Notice of Inquiry on 11-19-2019 based on the 4-30-2019 submittal. The revised emissions exceeded prior permit limits and the City was invited to Show Cause Hearing. The City attended the SCH on 12-13-2019. SCHD followed up with an Additional Information Request for Violation and Penalty Assessment on 1-7-2020.

The annual biogas usage by TVA along with the amount of biogas flared as waste gas by the Maxson Facility over the last 10 years is as follows (in millions of cubic feet):

Year	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
TVA	0	0	0	0	0	60	131	397	414	106
Flared	552	565	744	1167	1254	1094	896	688	574	746

The City received zero in revenue from TVA for the Maxson biogas for 2019.

Permit Violations – This year the Maxson Facility had 56 excursions of its NPDES permit limits related to treatment efficiency and performance. The excursions are described as follows:

In February, there were 5 permit excursions. There were monthly concentration and monthly pounds BOD excursions. There were monthly concentration, monthly pounds, and one weekly pounds TSS excursions. The monthly concentration and pounds excursions were thought to be caused by an influent of unusual character the first half of the month. The cause of the weekly pounds TSS excursion could not be determined.

In March, there were 16 permit excursions. There were monthly concentration, monthly pounds, one weekly concentration, three weekly pounds, and four daily maximum BOD excursions. There were monthly concentration, monthly pounds, three weekly pounds, and one daily maximum TSS excursions. The excursions were caused by a combination of high flows due to elevated Mississippi River stages, a slug industrial discharge, an influx of sand due to a sewer repair, and a broken primary sludge wasting line.

In June, there were 2 permit excursions. There was one daily maximum concentration BOD excursion and one daily maximum settleable solids excursion. The excursion was caused by a bulking clarifier during high influent flow due to rain fall.

In ~~June,~~ July there was one permit excursion. There was one daily maximum settleable solids excursion. This was caused by increased influent flow during a construction operation.

In October, there was one permit excursion. There was one monthly concentration BOD excursion. This was thought to be caused by an influent of usual character that occurred sporadically throughout the month.

In November, there were 3 permit excursions. There was one monthly concentration BOD excursion. There were one monthly concentration and one daily maximum TSS excursions. The causes of the excursions could not be determined.

In December, there were 28 permit excursions. There were monthly concentration, monthly pounds, two weekly concentration, two weekly pounds, and eleven daily maximum BOD excursions. There were monthly concentration, monthly pounds, one weekly concentration, and five daily maximum concentration TSS excursions. There were one daily maximum settleable solids and two dissolved oxygen excursions. This was caused by a blower malfunction and a pipe rupture below the aeration building.

The Maxson Facility has had 672 permit violations for treatment efficiency and performance since the issuance of state permits in 1982. The cumulative violations are shown in Figure 4 with Stiles Facility cumulative violations for comparison. The excursions are associated with different periods of plant history as summarized below:

Period	Average Number of Excursions Per Year	Description
1982-1985	30.5	First plant capacity expansion (Consent Decree)
1986-1994	4.0	Industrial loading increasing toward full capacity
1995-1997	24	2nd plant capacity expansion
1998-2010	4.4	Steady loading conditions
2011-2019	42.3	Collection system failures and major plant maintenance projects.

The Maxson Facility has had two years (1987 and 2010) with zero violations. The highest number was 2016 with 112 violations.

During 2011-2019, 71 of 385 violations at the Maxson Facility, or 18.4%, were for "pounds" permit limits violations. Beginning with the 2011 permit, the Stiles Plant no longer had to report pounds whereas the Maxson Facility continued having to report, creating a regulatory disparity between the treatment plants. The violations per year on a percentile basis since 1982 are summarized below:

Percentile	Number of Violations
0.25	3
0.5	8
0.625	15
0.7	23
0.75	26
0.9	53
0.95	74

Disinfection, Process Improvements, and Odor Control

In 2014, the disinfection, process capacity and treatment improvements, and odor control projects were combined into a single initiative called Construction Manager at Risk (CMAR). Treatment capacity increase was based on an average daily flow increase from 80 to 90 mgd at influent BOD concentration of 800 mg/l. The equivalent rated BOD capacity should be 600,000 ppd or greater once capacity increases and performance improvements have been documented.

CDM Smith continued construction activities related to Package 1 throughout the year, spending the majority of the time pouring concrete for the disinfection tank and performing electrical upgrades. A pre-construction meeting for Package 2 was held on 3-13-2019. The project consisted of installation of coarse bar screens and rehabilitation of interior concrete surfaces. The headworks bypass began full operation on 8-6-2019. The work involved removing part of the Screen Building roof to install the screens. The bypass was originally scheduled for 4 months. It was extended through 2-6-2020 to repair additional corrosion damage not covered in the engineer's report.

During the year, Package 2A covering secondary clarifiers, intermediate pumping improvements, and fine bubble diffusers advanced to 100% design and approved by SRF. Package 2B covering re-aeration tanks and a new blower building went through re-design and advanced to 90% design. Package 3 which covered bio-tower media and odor dispersion fans advanced to 90% design. Previously, it was separate Packages 3A and 3B.

Biogas Quality Investigation

Four (4) samples of biogas were collected throughout the year and analyzed for hydrogen sulfide and BTU. Hydrogen Sulfide concentrations averaged 1075 ppmv ranging from 120 to 2800. Energy values averaged 410 BTU/cubic feet (wet) ranging from 330 to 500. Average annual values of analytical results from previous years are summarized as follows:

Year	Soloxanes (ppbv)	Hydrogen Sulfide (ppmv)	BTUs/cubic foot (wet)
2012	250	8451	440

2013	124	19,820	468
2014	156	4591	441
2015	152	2836	483
2016	NA	935	320
2017	NA	1981	420
2018	NA	4540	350
2019	NA	1075	410

TVA Conversion to Natural Gas Related Projects

In August 2017, installation of the 5 new flares was completed by Kiewit Construction under TVA's contract and flaring of the waste gas using the new flares initiated. In September 2017, installation of the 4 bioreactors was completed and the reactors charged with media. Kiewit's contract to complete the job ended 11-5-2017 and operation turned over to the City. The City has continued to work with CDM Smith and Biorem to work out warranty issues related to construction as well as Biorem's obligation to achieve steady-state operation so that maximum hydrogen sulfide levels do not exceed 480 ppm per the TVA contract with the City.

In January, Standard Electric performed warranty work on the bioreactors under contract to Kiewit. CDM Smith found a hydrogen sulfide meter capable of reading elevated hydrogen sulfide levels up 5000 ppm. Lord & Company under subcontract to Kiewit completed necessary adjustments to controls and monitoring equipment. In February, Biorem and CDM Smith replaced probes and cleaned tanks. In March, Biorem inspected the media for sulfur accumulation. Maxson personnel completed insulating heat exchanger piping and the system was re-turned to service. Bioreactor #1 was functioning as intended, Bioreactor #4 was only marginal, and Bioreactors # 2 and #3 were not working at all.

In April, problems grew worse as the bioreactors continued operation. Calcium sulfate was observed forming in sumps and tearing up pumps and it was already time to replace the coalescing filter which is a very expensive item. Bioreactor #2 and #3 were subsequently removed from service and inspected by Biorem personnel. Large gaps in the media were observed in #2 and #3 indicating that they had never been fully charged. Spare sacks of media left on the premises by Kiewit were added (by use of a crane) but still more had to be ordered (from China) before the system would be considered fully charged. The system was then re-seeded using Maxson Facility return activated sludge and put back into service. The calibration of the City's hand held hydrogen sulfide meters was called into question by Biorem because of the consistently higher readings and more frequent calibration procedures were implemented by the City. By May, Bioreactor #1 was observed to be consistently achieving 400 ppm hydrogen sulfide discharge and Bioreactor #4 achieving 1000 ppm. In June, treatment was lost in Bioreactors #1 and #4 after a power failure shut down biogas production and bioreactor recirculation pumps. Biorem personnel removed sulfur from the system and re-seeded with biomass brought from another Biorem system in Illinois in an attempt to resume treatment.

In July, Biorem replaced all pumps and motors, replaced sprayers, and topped off the bioreactors with additional media and promised a "trip report" summarizing outstanding issues. A co-op student was then left on-site to monitor performance. Bioreactor #1 then shut down due to scale build-up. Both swings in biogas production related to changes in barometric pressure and inability to readily return to normal treatment after power failure impacted hydrogen sulfide removal efficiency. Despite these setbacks, Biorem visited the site off and on throughout the month of August in preparation for the required 30-day trial to determine if the system discharge could consistently meet TVA's requirements.

In September, the 30-day trial began with Biorem personnel on site for the majority of the month and the system hydrogen sulfide discharge was generally in the 100-200 ppm range. The conditions of the trial required that no equipment modifications be made. However, it was noticed that the discharge was out of specification in afternoon. The Biorem system succeeded at producing less than 100 ppm H2S for one continuous 24-hour period during the trial. A new in-line analyzer was installed at the beginning of the trial but the sensor failed due to high moisture content. The 30-day trial ended on 10-4-2019 with mixed results. A malfunctioning spray nozzle had to be replaced which violated the conditions of the trial.

In October, Biorem performed their last visit for the year. A replacement in-line hydrogen sulfide meter was installed but soon failed due to high moisture content of biogas and was returned to the manufacturer. The system appeared to be working satisfactory by mid-October based on hand held meter results. In December, City dedicated a wastewater treatment trainee to operate and be involved with system maintenance to provide consistency of operations. Maxson personnel replaced all piping with stainless steel and replaced PLC power supply.

The revenue contract was signed with TVA on 1-3-2018 with a 20 year term effective once biogas treatment is adequate. The estimated capital contribution is $20 million based on a 6,500,000 MMBtu minimum agreement volume with TVA concurrently agreeing to pay the City 33 cents per MMBtu spread out over a 13-year payback period. This is based on a minimum annual average 500 BTU wet (higher heating value) and maximum monthly average Hydrogen Sulfide concentration of 30 grains per 100 CF or 480 ppm.

TVA plans are to demolish the Allen Fossil Facility in about 2024. They plan to be dewatering ash ponds through at least 2026 and will be continuing to use their cooling channel for this discharge.

Biosolids Long-Term Planning

The Maxson Facility began operations in March 1976. Lagoon #1 was constructed with the main plant in 1975. Lagoon #2 was constructed as a supernatant clarifier in 1977-78 with supernatant returning to the head of the plant. Lagoons #3, #4, and #5 lagoons were constructed during this same time period. TDHE approved the 160 acres south of Lagoon #1 for lime stabilization. The City sent a draft interim sludge handling plan to TDHE on 5-11-1978. The Consent Decree entered on 7-28-1978 had a section requiring the City to initiate an interim solution for "sludge handling" facilities. Pursuant to the Decree, the City built an 8-acre pond (Lagoon #3) and proposed to spray-irrigate the sludge on an adjoining field in the dry weather months. On 12-1-1978, the City began discharging sludge to Lagoon #3 which was anticipated to provide storage capacity for one year. As of 4-10-1979, the lagoon was filled to capacity and sludge began to flow back through the plant. The Second Amendment to the Consent Decree was entered on 5-7-1979 to allow the City to intermittently discharge sludge through the plant's bypass until there was sufficient capacity to store sludge until startup of the land application program later that year. A revised sludge handling plan was submitted to TDHE on 7-2-1979. This plan stated that lime stabilized sludge would be applied to the 110 acres just south of the anaerobic lagoon during summer of 1979 with approximately 350 acres of additional land available adjacent this property for future use. On 1-2-1980, the City submitted a final revision of the plan that focused on sludge land application. By 5-6-1980, the City received approval for a sludge spray irrigation system on the 110 acres (west of the lagoons) with an additional 310 acres available for future use. The City acquired two sludge trucks with 4000 gallon tanks in May of 1980 and received a permit to apply sludge to a 310 acre dedicated site adjacent to the lagoons in June of 1980. By that point, a total of 280 million gallons of sludge storage capacity was available for stabilization and further digestion, followed by application on approximately 420 acres of dedicated land.

The City's sludge management initiative came into being against an evolving backdrop of state and federal regulatory programs. Sludge reports submitted to TDHE from 1980 to 1992 documented quantity of sludge applied and analytical results and as time went on, surface water run-off analytical results. Four sludge application sites were approved for the Maxson Facility beginning with Site 1 (110 acres) in 1979, followed by Site 2 (310 acres), in 1980, Site 3 (180 acres) in 1983, and Site 4 (3500+ acres) in 1982. Most of the sludge was applied to Sites 1, 2 and 3 at a rate of 15 to 20 dry tons per acre per year in an area that became known as the Earth Complex. Site 4 consisted of available agricultural land in Pidgeon Industrial Park where sludge would be applied at lower rates as fertilizer but apparently never extended beyond a nearby 500 acre site leased by a farmer. In September 1987, TDEC transferred all regulatory authority for municipal sludge disposal facilities from SWM to WPC in accordance with the Federal Water Quality Act of 1987 amendments. New NPDES permits effective 1-29-1993 incorporated annual and monthly reporting requirements for sludge management practices under Other Requirements.

The Maxson Facility added three primary clarifiers that became functional in 1982, resulting in the production of primary sludge. Initially, the primary sludge was sent to Lagoon #1. Sludge handling facilities for primary sludge were constructed and started up during the summer of 1983. These facilities included three belt filter presses with a total capacity of about 100 dry tons per day and mixing with lime for stabilization. The dewatered primary sludge was then land applied to Site 3 beginning about 1984. This practice continued until April 2004 when all primary sludge began being wasted to the lagoons.

Digested sludge disposal continued through 1993 by spreading, injecting or spraying liquid 1-2% sludge onto the fields by tank truck. New federal regulations known as 40 CFR 503 became effective 2-19-1994 which included Surface Disposal of sludge that met appropriate vector attraction reduction and metals criteria. The City opted to comply with federal Surface Disposal Site requirements that included containment of surface run-off and ground water monitoring and so were no longer regulated by the state program. Berms and ditches were constructed to direct surface run-off and supernatant from the approximately 600 acres of dedicated land into the Nonconnah Interceptor or via a gated 18-inch line leading into a manhole on the 84-inch Horn Lake Interceptor. Lagoon #1 lagoon was covered 1995 and the flaring of biogas began. Secondary Dewatering was placed in operation 1996 with sludge surface disposed in the new dedicated site. The use of the sludge spreading equipment was gradually phased out and replaced with dump trucks and dozers over the next several years.

An anaerobic digestion study was done in 1998 to optimize current lagoon operations, including the wasting of primary sludge into the lagoons which, at this point, had rarely been done. Design of new covered Lagoons #2A, #2B, and #2C soon followed and constructed in 2001-2003. Plans were made to cover Lagoon #5 at a later time. Sale of biogas to TVA began 2000 based on a 20-

year agreement. The new covered lagoons were placed in service in spring of 2004 and the plan to waste primary sludge to the lagoons was enacted, ushering in a 12 year period of double biogas production. Unfortunately, TVA did not properly maintain their equipment and were unable to utilize a great deal of the biogas the City produced. Lagoon #1 cover was replaced in 2011 due to cover failures dating back to 2007. Figure 5 showing long-term accumulation of Maxson Facility surfaced disposed biosolids and sludge accumulation within the lagoons. As shown, the new Secondary Dewatering practices implemented in the mid-1990s were able to hold down lagoon sludge accumulation through about 2003. Beginning with 2004 the rate of accumulation began increasing. This is despite the refurbishment of belt filter presses (2005), the purchase of three new additional belt filter presses (2013), and the replacement of the two lagoon dredges with new units (2013). Hopefully, the addition of a third dredge purchased in 2019 to mine sludge from further out in Lagoon #5 will increase the rate of surface disposal.

Operational Impact of Iron Salts

Ferrous chloride addition in large quantities began December 1998 in order to control hydrogen sulfide that was prevalent in the new Secondary Dewatering facility adding a considerable amount of iron to the lagoon system over the years. All ferrous chloride deliveries halted January 2012 due to EPA concerns regarding the practice the City employed to receive the ferrous chloride by unloading directly into the lagoons from tanker trucks. A summary table of yearly iron salts received at the lagoons versus land applied sludge is provided below:

Year	Irons (million dry pounds) added to lagoons	Surface Disposed million dry pounds
1999	5.35	45
2000	4.6	45
2001	2.78	48
2002	4	49
2003	3.7	40
2004	5.06	37
2005	3.85	35
2006	2	29
2007	2.1	51
2008	2	78
2009	1.2	45
2010	3.1	56
2011	4.1	42
2012	0	24
2013	0	31
2014	0	23
2015	0	27
2016	0	30
2017	0	25
2018	0	23
2019	0	16

The impact of iron salts appears to be cumulative since hydrogen sulfide vapors at Secondary Dewatering are still under control these many years later. However, as shown above, the discontinuation of iron salt usage possibly contributed to the increasing inability to find good feed solids over the last several years.

It is believed that iron salts added to the headworks when the plant was primarily in roughing played a role in maintaining the stability of this process and the subsequent instability of this process when the practice was discontinued as shown below:

Year	Effluent BOD mg/l	% Roughing Mode	Iron Added to Headworks, ppd	% BOD removal
2007	26	100%	10,685	97%
2008	34	100%	6301	96%
2009	24	100%	2014	96%
2010	20	100%	7123	97%
2011	31	66%	5753	96%
2012	39	25%	0	95%
2013	36	92%	0	96%
2014	34	17%	0	96%

Mississippi River Impact

The Maxson Facility does not sustain major flow impact until the Mississippi River stage crosses 38 feet. Figure 6 shows the impact of river stage on average daily flow after about a week period of saturation above 38 feet. This same phenomenon was previously reported in the 1997 Annual Report.

2.0 Stiles Wastewater Treatment Facility

Introduction- On 1-7-2019, the wet well level jumped from 192.5 to 205.8 feet as the Mississippi River stage crossed from 30.29 to 31.14 feet, thus ushering in the longest continuous period of high river stage and influent flow in Stiles Facility operational history.

A kickoff meeting was held on 2-14-2019 for the Lagoon 3N design project to increase the outfall from 10-inch to 18-inch diameter. TDEC held a public meeting on 2-26-2019 for the Stiles draft NPDES permit and granted an extension to receive public comments to 3-13-2019. TDEC would not impose a permit limit on PAA residual since the City would have an incentive to keep it low due to cost.

As the Mississippi River stage crossed above 40 feet on 3-1-2019, there was a power failure due to an electrical fault between one of the blower transformers and the OCB yard. This caused an upstream manhole on the Loosahatchie Interceptor adjacent Lagoon #3 to blow off. Power was quickly restored by electricians already working 12-hour shifts on-site. The remaining electrical feeder, however, was snapping and popping under the strain. Emergency backup generators were brought in for the next several months as the high river stages continued. The Phase 2 Electrical Upgrade project as outlined in 2014 was expedited starting with an emergency engineering and constructions services with Pat Jorgensen's firm and Tri State to replace six feeder cables and provide three spare cables stored at the Environmental Maintenance facility at 8245 Truitt which will have to be tracked as off-site inventory. The construction cost was $1,345,233 including equipment and materials and the backup generator rental was $960,000. On 3-6-2019, the Stiles Facility Influent Pump Station Joy Fan Room floor began sustaining leaks as the river stage crested at 41.1 feet (231 feet elevation). The Joy Fan Room floor elevation is approximately 225 feet so the floor was being subjected to substantial head pressure. Plant maintenance personnel installed a pressure plate to control the leak. On 3-22-2019, a raw sewage pump momentarily shut down due to water getting into oil, reducing pumping capacity. The river stage began falling over the next several days lowering through put requirements but reducing pump efficiency. Despite decreased pump efficiency, the pumps were able to maintain wet well below the critical 216-217 feet elevation necessary to keep manholes in the immediate vicinity of the plant from overflowing.

In June, the new pre-final effluent sampling system installed as designed had too much entrained air based on original flow meter insertion points installed by CDM Smith back in 2014 so it was back to the drawing board. A lightning storm on 6-27-2019 resulted in decoupling a 2-inch influent sampling line going to the wet well that had become loose due to vibration. Bids came in for the lab renovation and the lowest bid was about $3.2 million.

There was a power outage on 7-9-2019 due to a snake getting into a power circuit. Barge Design conducted a workshop on 7-10-2019, focusing on improving the reliability of the influent pump station. On 7-12-2019, the Mississippi River Stage finally crossed below 30 feet, above where it had spent the majority of time since the first of the year. Average daily flow returned to typical dry weather conditions around 7-26-2019 for the remainder of the year.

Barge Design submitted their secondary clarifier evaluation report on 8-12-2019 summarizing work performed the previous year. The Stiles Facility received a Notice of Violation on 8-20-2019 for Significant Non-Compliance from December 2018 through June 2019. TDEC has been criticized in the past by EPA for not flagging DMRs and issuing NOVs using SNC review criteria. On 9-25-2019, the City transmitted the Corrective Action Plan to TDEC in response to the NOV issued on 4-18-2018 for the sanitary sewer discharge / bypass into the Mississippi River.

The final walk-through for the Stiles Disinfection Facility was performed on 9-16-2019. Additional work included cutting slots in the vertical baffles at the water line to allow scum and other floatables to pass through to the scum removal trough adjacent the overflow weir, a condition which becomes more problematic in the colder months.

Additional fibers have been noted in the effluent and occasionally influent grab samples in the past including a slug discharge on 10-29-2019. This corresponded with a shut down across the street at KTG as they ran out their "stock chest."

PXC Wolf River PAA production plant was commissioned on 12-18-2019. However, use of the buried product transfer line was delayed due to failure to pass leak testing.

The Maintenance Department performed numerous noteworthy tasks during the year. In January, a new Panel Alarm system was installed on #2 and #4 Blower at the Blower Building. In February, a new motor and fan was installed in the East Fan section of the Joy Fan Room in the Pump Building. In March, the north end corner seam of the Return Sludge Channel wall where it was pulling away from the Raw Sewage Influent Channel was repaired. At Chapel Hill, new fencing was installed around the perimeter. A new Electrical Distribution System was installed for the boiler at Old Dewatering.

In April, all exterior doors of the Administration, Maintenance, and Pump Buildings were repainted. Both 20 MVA Transformers were rehabbed, fins removed and gaskets replaced at the Blower Building and new cooling fans installed, all leaks fixed, and coils tested. A new mag meter was installed to meter the discharge of American Yeast ground water going into the Disinfection Tank. In May, chutes were redesigned and fabricated for the Fine Bar Screen Wash Press. #4 Raw Sewage Motor and Mag Drive units were reconditioned. In June, crews installed a new Westech Drive in place of old Singer Drive at A- 2 Final Tank. The Joy Fan in Blower #4 Bay was replaced at the Blower Building. A new duct bank and feeders for both transformers were installed by a contractor into the Blower Building. Staff also installed a new stainless airline system throughout New Dewatering and replaced old air lines.

In July, crews installed new Air Compressor system in Old Dewatering to replace Air Compressors in the Electrical Room. In August, maintenance staff installed new LED High Bay Light Fixtures to replace old Metal Halide Fixtures in the Old Dewatering Truck Bay. Crews installed a new boiler in Old Dewatering Building to provide heat. In September, new baffle walls were installed in the tanks at Chapel Hill.

In October, crews rebuilt #4 Raw Sewage Pump. In November, crews installed a second J-line submersible pump in the Pump Building dry well to have extra emergency redundancy. Staff sealed and coated floors and walls to control leakage in the Pump Building Joy Fan Room. In December, Grit Tank #2 Drag-Out was completely rebuilt and new ¾-inch floor plate installed. All outdated Fluorescent Lighting Fixtures in the Admin Building were replaced with LED fixtures. A new pressure washing system was installed in the Pump Building. Epoxy was injected in cracks in the walls of the stairwell to stop seepage from into the Wet Well into the Dry Well stairwell.

In addition to these major tasks, the Maintenance Department completed 3178 Preventative Maintenance Work Orders, 531Scheduled Maintenance Work Orders, and 239 Unscheduled Maintenance Work Orders for a combined total of 4013 Work Orders (including some not categorized). The annual number of work orders completed over the last 10 years is summarized as follows:

Year	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
Preventative	3178	2795	3148	2694	2738	2802	2832	2402	2563	2741
Scheduled	531	758	497	345	552	799	837	1049	781	941
Unscheduled	239	255	188	138	156	192	323	364	325	432
Total	4013	3863	3857	3177	3446	3810	3992	3747	3669	4114

Flow – The average daily influent flow for the Stiles Facility in 2019 was 113 MGD. This is an increase of 34 MGD from the previous year. The 0.5th percentile flow in 2019 was 60 mgd which is considered base flow for the year. The average daily flow from the major significant industrial users was 14.2 mgd. This leaves 38.8 mgd which is 34% of average daily flow unaccounted for. The rainfall was 71.5 inches (plant gage). The average annual rainfall at the Stiles Plant gage has been 55 inches.

BOD_5 – The average influent BOD_5 in 2019 was 265,315 pounds per day. This is 2,799 pounds or 1% less loading from the previous year. The average influent BOD_5 concentration was 313 mg/l and the average effluent concentration was 59 mg/l for a BOD_5 removal efficiency of 81%. The average effluent BOD concentration adjusted for PAA was 52 mg/l for a BOD removal efficiency of 83%. The total average BOD_5 in pounds per day reported discharged by the major significant industrial users was 174,582 pounds per day in 2019 which equals to 66% of the total loading. This is 4,803 pounds or 3% more than the previous year. The 2nd percentile loading in 2019 was 118,504 ppd which is estimated domestic loading for this year. This leaves all influent BOD loading more than accounted for.

Total Suspended Solids (TSS) – The influent total suspended solids averaged 295,671 pounds per day in 2019. This is 1,156 pounds or <1% more than the previous year. The average influent TSS concentration was 343 mg/l and the average effluent concentration was 40 mg/l for a TSS removal efficiency of 88%. The total average TSS reported discharged by the major significant industrial users was 62,837 pounds per day in 2019 which equals to 21% of the total loading. This is 1,860 pounds or 3% less from the previous year. TSS loading from lagoon supernatant and dewatering filtrate return was calculated to be 28,359 ppd or 9.6% of total average daily TSS loading. The 2nd percentile loading for 2019 was 118,864 ppd which is estimated domestic loading for this year. This leaves 85,611 ppd of TSS or 29% of average daily loading unaccounted for.

Chemical Oxygen Demand (COD) - The influent averaged 764 mg/l and 643,019 ppd. The effluent averaged 238 mg/l and 224,296 ppd that include PAA. The 2nd percentile loading for 2019 was 246,466 ppd which is estimated domestic loading for this year. The total average COD discharged by the major significant industrial users was 382,236 ppd which equals 59% of the total loading. This leaves the remaining 14,317 ppd or 2.2% of influent COD loading unaccounted for.

Effluent E. Coli - Effluent E. coli analysis was performed on daily effluent grab samples throughout the year as required by the NPDES permit as report only. Effluent E. coli results for this past year are presented in Figure 7 based on disinfected effluent.

Water – In 2019 the metered water usage totaled 333,070 CCF at a cost of $341,066. The monthly average metered water consumption was 33,307 CCF or 0.68 MGD. This is a decrease of 2041 CCF ~~per day~~ from the previous year. A project was implemented in November 2019 to meter diverted well water from American Yeast so that a complete accounting of all water use can be made that will serve as a baseline for future water conservation efforts. Flows discharged directly from American Yeast for use by the Stiles Facility (i.e. aeration basin and scum hopper spray) averaged 1.7 MGD.

Energy – The plant main power consumption was an average of 5,575,198 kwh of electricity per month in 2019. This is approximately 106,420 kwhs per month more than the amount used in the previous year. The monthly peak demand averaged 8581 kw which is 106 kw more than the previous year. The average monthly main power bill in 2019 was $384,235. This is $7479 less per month than the previous year. Payments for participating in the ENERNOC curtailment program for main power totaled $40,496.

Both August and October were the high month for demand at 9072 kw. January was the high month for consumption at 6,038,418 kwh. March was the low month for demand at 6981 kw and May was the low month for consumption at 4,918,711 kwh. Daily power consumption throughout 2019 is shown in Figure 8.

Effective 10-1-2015, the City entered into an agreement with MLGW to switch from Seasonal Demand to Time of Use billing. TVA would charge Fuel Cost Adjustment based on user class. TOU rate structure benefits high load factor customers through incentivizing off-peak energy usage. The contract term is 5 years and will be automatically extended unless otherwise terminated. The City signed an alternate circuit agreement with MLGW on 10-2-2019 for 10 years for the circuit switches to be on the MLGW power poles. This will result in an additional $13,778.23 monthly charge on the main power bill once the program is in place.

Sludge and Biosolids – The Stiles Plant wasted 101 million dry pounds of secondary sludge to the lagoons in 2019. This is 13.7 million pounds less compared to the previous year. 35.3 million dry pounds (based on counting trucks) was surface disposed at a 91% capture rate.

The annual amount of digested sludge (in million dry pounds) removed from the two covered lagoons over the last 10 years is provided below:

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
38.8	43.9	31	44	30	29.4	29	36	17.6	34

Approximately 27% of the secondary sludge wasted to the lagoons in 2019 was destroyed. The formula used to calculate percent destroyed is %VS of WAS multiplied by %VAR. This equals

28 million dry pounds (13,928 tons). Considering land filling and volatilization the new change in the lagoon solids inventory was an increase of 38 million dry pounds within Lagoon Nos. 1, 2, and 3. The calculated cumulative amount of dry solids within the three lagoons is 725 million dry pounds. In 2019, approximately 35 million dry pounds (17,663 dry tons) of bio-solids were disposed in the Surface Disposal Site. The calculated amount of dry bio-solids that has been surface disposed since 1984 is 1346 million dry pounds.

For the last 10 years, net accumulation has been occurring at an average rate 27.48 million dry pounds per year. Digested sludge removal from the two covered lagoons has averaged 33.38 million dry pounds per year. In order to stop net accumulation, the Stiles Facility will need to dewater and surface dispose or land apply an average of 60.86 million dry pounds per year which would be nearly double the existing rate and a 98 percentile level of performance.

The total cost for the dewatering and surface disposal operation was approximately $3,282,320. This equates to a unit cost of $104 per dry ton of bio-solids surfaced disposed and volatilized (combined) in 2019. The annual cost per dry ton for surface disposal and volatilization combined over the past 10 years is as follows:

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
$104	$83	$93	$85	$102	$70	$78	$67	$57	$86

Generally, the unit cost to dewater and dispose sludge as bio-solids is a function of polymer cost and dewatering throughput. The cost to dewater and surface dispose digested sludge was $186 per dry ton in 2019. The annual cost per dry ton based on dewatering throughput over the past 10 years is as follows:

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
$186	$169	$250	$168	$129	$199	$192	$162	$189	$129

Biogas- Approximately 135.6 million cubic feet of biogas was produced 2019. The annual biogas production over the last 10 years is as follows (in millions of cubic feet per year):

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
135.6	111.7	121.7	105.9	158.7	275	263.8	338.7	272.5	350

The annual daily average bio-gas production over the last 10 years is as follows (in million cubic feet per day):

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
0.37	0.31	0.33	0.29	0.43	0.75	0.72	0.93	0.75	0.96

The biogas produced equates to 4.86 cubic feet of biogas per pound of volatiles destroyed. The yield (ratio of biogas to pounds of volatiles destroyed) during the 10-year period is shown below:

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
4.9	2.8	2.8	2.7	4.7	8.3	9.9	10.1	10	10.7

Like the Maxson Facility, the amount of sludge volatilized each year is relatively constant. One would expect the biogas yield to be constant but it is not as shown above. What is variable is the volume of biogas produced. It could be that a portion of the sludge is not actually being destroyed but is accumulating as a stabilized volatile fraction. This stabilized volatile fraction is not being properly mixed and broken up to allow anaerobic microbes to digest it due to short-circuiting and the reduction in viable lagoon volume. Trends in biogas production over the years are shown in Figure 9.

In 2019, American Yeast received 48.87 million cubic feet of biogas from the Stiles Facility which is nearly 36% of total biogas production. The revenue received from American Yeast for this biogas was $36,815. The annual usage by American Yeast for the last 10 years along with the amount of biogas flared as waste gas by the Stiles Facility is as follows (in millions of cubic feet):

Year	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
American Yeast	48.9	71.9	85.3	63.1	80.5	84.3	99.35	105.7	76.2	109.6
Flared	90	39.8	36.4	42.7	78.2	191	164	233	196	240

Permit Violations- This year the Stiles Facility had a 181 excursions of its NPDES Permit limits related to treatment efficiency and performance.

In February, there were 2 permit excursions. There were monthly average percent removal and average concentration exceedances for BOD. This was caused by excessive influent flow due to high Mississippi River stage.

In March, there were 42 permit excursions. There were ten daily minimum percent removal, seventeen daily maximum, four weekly average concentration, monthly average percent removal, and monthly average concentration exceedances for BOD. There were six daily maximum, monthly average percent removal, and monthly average concentration exceedances for TSS. There was one bypass of treatment violation due to mixed-liquor overflowing the aeration basins. These were caused by excessive influent flow due to high Mississippi River stage.

In April, there were 21 permit excursions. There were one daily minimum percent removal, thirteen daily maximum concentration, three weekly average concentration, monthly average percent removal, and monthly average concentration BOD exceedances. There were monthly average percent removal and monthly average concentration TSS excursions. These were caused by excessive influent flow due to high Mississippi River stage.

In May, there were 34 permit excursions. There were five daily minimum percent removal, twenty-one maximum daily concentration, four weekly average concentration, monthly average percent removal, and monthly average concentration BOD exceedances. There were monthly average percent removal and average concentration exceedances for TSS. These were caused by excessive influent flow due to high Mississippi River stage.

In June, there were 33 permit excursions. There were four daily minimum percent removal, twelve maximum daily concentration, four weekly average concentration, monthly average percent removal, and monthly average concentration exceedances for BOD. There were nine daily minimum percent removal, monthly average percent removal, and monthly average concentration exceedances for TSS. These were caused by excessive influent flow due to high Mississippi River stage.

In July, there were 39 permit excursions. There were two daily minimum percent removal, twenty-two daily maximum concentration, four weekly average concentration, monthly percent removal, and monthly average concentration exceedances for BOD. There were six daily minimum percent removal, one daily maximum concentration, monthly average percent removal, and monthly average concentration exceedances for TSS. These were caused by excessive influent flow due to high Mississippi River stage.

percent removals

In August, there were 10 permit excursions. There were six daily maximum, two weekly average concentration and monthly average concentration exceedances for BOD. There was one weekly average concentration exceedance for TSS. These were caused by excessive influent flow due to high Mississippi River stage.

The Stiles Facility has had 990 permit violations for treatment efficiency and performance since the state permits were issued in 1982. The cumulative violations are shown in Figure 4 with Maxson Facility cumulative violations for comparison. The excursions are associated with different periods of plant history as summarized below:

Period	Average Number of Excursions Per Year	Description
1982-1984	106.7	Commissioner's Order
1985-1992	6.5	Municipal Compliance Plan Annual Reporting
1993-2010	14.5	Occasional collection system failures and high river stages
2011-2019	39.7	Increasing collection system failure and high river stages.

The Stiles Facility has had 5 years (1987, 1992, 1999, 2001, and 2007) with zero violations. Excluding years of high river stage and/or major collection system failure, the Stiles Facility has averaged 2.5 violations per year (22 out of 38 years). The highest number on record was 2019 with 181 violations.

The pounds-based parameters were eliminated from the Stiles Facility permit beginning in 2011 with only concentration and percent removal parameters remaining for BOD and TSS. The Maxson Facility permit, however, retained the pound parameters. During 2011-2019, 18.4% of the violations of the Maxson Facility were pounds-based. The Stiles Facility number of violations potentially could have been higher had the pounds permit parameter been retained.

The violations per year on a percentile basis since 1982 are summarized below with Maxson Facility for comparison:

Percentile	Stiles	Maxson
0.25	1	3
0.50	7	8
0.60	12	12
0.68	23	23
0.70	27	23
0.75	39	26
0.80	40	29
0.85	58	36
0.90	100	53
0.95	154	74

As shown above, the rate of violations between the plants is similar up to the 68th percentile level. Beyond this level, the Stiles Facility has more violations compared to the Maxson Facility escalating to twice as many beyond 90th percentile, even with the recent advantage of having pounds eliminated as a permit parameter.

Chapel Hill

At the Chapel Hill package treatment plant, there were 12 NPDES permit limit excursions. A new permit was issued 7-1-2017, effective 8-1-2017, and expiring on 7-31-2022. The City took over operation and maintenance of the Chapel Hill package treatment plant in 1972 by Council Resolution. The property was deeded from the developer to the City of Memphis on 6-28-1972, including the sewer mains and laterals. The City expends approximately $110,000 annually on operation and maintenance of that package plant. Planning was initiated in 2014 to either upgrade or replace the package plant as it is showing signs of wear. On 2-9-2015, Fisher & Arnold submitted their final report which found that the area is not in anyone's annexation reserve, the facility was basically structurally sound, the bio-towers need to be replaced and the concrete basin rehabbed. Early in 2018, the Chapel Hill influent pump and blower were connected to the Mission system by cell phone to allow remote monitoring of operations of raw sewage pumping and aeration. Discussions were initiated in 2017 to transfer operation and ownership of the package plant to Shelby County after substantial improvements have been made. The improvements are expected to be made in 2021 as a capital improvement project and if everything goes as planned, the facility will be transferred to Shelby County.

Bio-solids Long Term Planning

The current configuration of anaerobic digestion in covered lagoons, dredging and pumping to belt filter presses for dewatering to 15% solids, followed by solar drying to 50% solids in a dedicated surface disposal site a short distance away represents a very cost-effective economic model since all of this is operated and maintained in a routine manner by City personnel. The scale, reliability and effectiveness of this operation is impressive. Typically rainfall occurs at least 70 days per year in the Memphis area which impacts spreading and solar drying.

Under Pathogen Reduction for Surface Disposal, the plant has been reporting under Class B, Process to Further Reduce Pathogens, Alternative 2 (anaerobic digestion). Under Vector Attraction Reduction, the plant has reported under Alternatives 1 and 2 (greater than 38% volatile reduction, less than 17% additional volatile reduction) based on waste activated sludge as the starting point and feed solids as the end point for Alternative 1. Prior to 2002, dewatered cake was the end point for Alternative 2. Beginning in 2002, feed solids were the end point for Alternative 2.

For some historical perspective, the Stiles Facility generated 245 million dry pounds of sludge from 1978 to 1982 after the treatment plant first opened. In 1980, the City designated a 53 acre area adjoining the south end of the airport runway as a temporary bio-solids landfill known in the early planning documents as Sludge Landfill No. 1. A scope of operation plan dated 4-14-1981 was prepared for this facility and approved by the TDPH. During 1980 to 1983, the plant began dipping sludge out of the lagoons and disposing it in Sludge Landfill No. 1 of which only 35 acres was actually used. There was no dewatering performed at that time as the original dewatering facility was improperly designed so a second contract was issued to build a new facility. Lagoons No. 3 and No.4 comprising 65 acres were permitted by TDHE as a sewage sludge landfill on 1-18-1984 with the requirements that groundwater monitoring wells be installed and that runoff does not constitute a threat to the environment or public health. This was known as Sludge Landfill No. 2. The City constructed Lagoon No. 4 and began using it in the summer of 1984. New federal regulations known as 40 CFR 503 became effective 2-19-1994 which included Surface Disposal of sludge that met appropriate vector attraction reduction and metals criteria. The City opted to comply with federal Surface Disposal Site requirements that included containment of surface run-off and ground water monitoring which the City was currently doing and so was no longer regulated by the state program. This became established as the Surface Disposal Site for the next 25+ years.

Figure 10 shows the calculated cumulative net change in material within Lagoon Nos. 1, 2, and 3 and the cumulative amount of dry solids stored in the Surface Disposal Site. The Surface Disposal Site is nearing capacity and it is becoming more of a challenge to operate due to increasingly poor surface drainage.

During January-May 2019, Peter Alfonso collected 280 samples from Lagoon Nos. 1 and 2 and submitted for %TS and %TVS analysis. 89 of those samples were also submitted for metals, nutrients, sulfides, and fecal coliform. The study determined that Lagoon No. 1 contained 37,350 dry tons and 32,664 dry tons. Lagoon No. 1 had average TS = 9.6% and VS = 51.3%. Lagoon No. 2 had average TS = 8.4% and VS = 52.78%. All of the samples analyzed for metals, with two exceptions, were below 503 limits. TKN ranged from 6980 to 106,000 ppm (dry). Phosphorus ranged from 5850 to 31,100 ppm (dry). Sulfides were detected ranging from 368 to 4500 ppm (dry) indicating a potential odor concern if exposed to the atmosphere. Fecal Coliform ranged from 17 to 333,000 MPN per gram-dry. A summary of the quantity of sludge storage in Lagoon Nos. 1, 2, and 3 based on in-situ investigation is provided below:

	Sludge Quantity (dry tons)	Time Period Determined
Lagoon 1	37,350	January – May 2019
Lagoon 2	36,664	January – May 2019
Lagoon 3S	109,926	November – December 2016

Lagoon 3N	102,500 (estimated)	8-30-2019 RFP
Total	286,440	

Based on record keeping and volatile reduction calculations, the amount of material stored in the three lagoons is 362,000 dry tons. An additional 18-20% reduction of the original material may have occurred due to longer term volatilization given the extreme 35+ year holding period. This would put the in-situ quantity based on record keeping at 289,600 dry tons. Lagoon 3S is believed to be near sludge storage capacity based on an investigation completed during the months of November and early December 2016 with an average total solids content of 15%. Lagoon Nos. 1 and 2 are useable as long as the City can reduce the net accumulation rate. The City continued drying out Lagoon 3N using heavy equipment in order to gain use of the area as a future surface disposal site and used it on occasion for additional surface disposal due to worsening surface conditions in the Surface Disposal Site.

A workshop was facilitated by Barge Design on 2-4-2019. The meeting consisted of a review of suggestions to improve dewatering performance for the upcoming biosolids land application project as well as on-going influent pump station evaluation, collection system inflow-infiltration analysis based on system flow metering, secondary clarifier evaluation, and a preliminary engineering evaluation for dewatering improvements studies. On 3-22-2019, Barge Design conducted an update workshop the 30% Design / Preliminary Engineering Review for sludge management, American Yeast beer waste co-digestion, and flow monitoring study to determine EQ basin storage requirements to reduce peak rain-induced flow. An RFP for Class B biosolids land-application was issued on 5-7-2019 and a pre-proposal meeting was held with prospective proposers on 5-15-2019. Interviews were performed with the top two proposers on 10-8-2019 and Denali was subsequently selected with a bid price of $29 per wet ton for Class B land application. Barge presented their final 30% Design / PER for sludge management / solids processing upgrades on 10-8-2019. Total cost range is $110 to $121 million.

Disinfection and PAA Impact on Final Effluent BOD

Continuous PAA dosing began on 11-10-2018. The PAA usage for 2019 and calculated dosage of active ingredient is provided as follows:

Month	Amount of 15% PAA Solution Used (pounds)	Calculated Average Dose of Active Ingredient (mg/l)	Increase in Soluble BOD Between Pre-Final and Final Effluent (mg/l)	Final Effluent BOD (mg/l)	Adjusted Final Effluent BOD (mg/l)
January	1,638,000	9	3.0	37	34
February	1,761,000	9	7.4	45	38
March	2,124,000	7	2.3	85	83
April	2,580,000	10	6.6	83	76
May	3,276,000	12	-3.7	93	97
June	2,715,000	11	3.5	78	75
July	3,885,000	19	22.4	105	83
August	2,424,000	19	10.4	57	47
September	1,920,000	16	7.2	30	23
October	1,878,000	14	5.6	30	24

November	1,707,836	14	9.0	34	25
December	1,616,055	9	9.3	35	26
Average	2,293,741	13	6.9	59	53

Sampling performed after initiating PAA disinfection showed an increase in soluble BOD downstream of the dosing point compared to the upstream sampling location. This increase in soluble BOD is shown in the above table as well as the calculated Final Effluent BOD adjusting for PAA impact. Since disinfection began in November 2018, Soluble BOD correlates with PAA dose at an $R^2 = 0.56$ based on monthly average increase of sBOD = 0.52 x PAA Dose (mg/l) excluding two outliers (May 2019 and July 2019) as shown in Figure 11.

PXC's algorithm consists of a base dose and a color related dose. PXC was challenged for the first seven month of the year by elevated flows related to high river stages. This required manual adjustment of the base dose to maintain a PAA residual. During the last 5 months of the year, PXC was challenged by more concentrated color and/or color of a different nature in the pre-disinfected effluent. The below statistics were calculated based on PXC invoices and City generated data, including pre-final effluent color concentration.

Condition	ADF mgd	Base PAA usage ppd	Calculated Base Dose mg/l	Color PAA Usage ppd	Calculated Color PAA Dose mg/l	Total PAA avg dose mg/l	Number of Daily Max E coli Violations	Months
Elevated Flow	140	48,073	6.4	37,059	4.9	11.3	0	7
Normal Flow	74	4,587	1.2	57,780	14.6	15.7	9	5

As shown above, during elevated flow conditions associated with high river stages, 43% of the dosage was related to changes in color and 57% related to changes in base dosage. During normal flows, 93% of the dosage was related to changes in color and 7 % related to changes in base dosage. There were no daily maximum E coli permit exceedances during elevated flow conditions and 9 daily maximum E coli permit exceedances during normal flow conditions, indicating more fine-tuning of the dosing is in order. It remains to be determined the degree of overdosing, if any, beyond an acceptable limit, that may have occurred during elevated flow conditions.

The other major differentiator in PAA dosage is whether or not Memphis Cellulose is on line. The following statistics were derived for various parameters during 2019:

Parameter	Memphis Cellulose Down	Memphis Cellulose Running	p-value	Overall Average
PAA Solution Usage ppd	66,473	80,504		75,623
PAA mg/l	11.6	13.7	0.119	13.0

Influent COD ppd	603,241	659,254	0.000	640,583
Pre-final Effluent COD ppd	223,334	252,487	0.333	242,976
Pre-final Effluent Color conc	604	789	0.001	726
Pre-final Effluent COD mg/l	239	268	0.040	258
Influent Flow mgd	106	109	0.242	108
E coli (max)	5200	8100		
E coli (geomean)	55	52		53
Number of Daily E coli Permit Exceedances	3	6		9 (total)
Color:COD Ratio	2.81	3.21	0.00	3.08
Pre-final Effluent sBOD mg/l	29	39	0.026	36
Effluent BOD	50	64	0.015	59
Increase in sBOD mg/l	6.86	7.03	0.097	6.97

The above statistics reveal a lot of useful information. When Memphis Cellulose is on-line, for example, influent COD loading increases by 56,013 ppd. Approximately 50% of this COD appears as additional COD in pre-final effluent (prior to disinfection). This 12% increase in pre-final COD concentration is associated with a 30% increase in pre-final color concentration. Another observation is that when Memphis Cellulose is on-line, pre-final sBOD increases from 29 to 39 mg/l and final effluent BOD increases from 50 to 64 mg/l. This indicates that a minimum of 4 mg/l of the increase in final effluent BOD when Memphis Cellulose is on-line could be attributed to PAA.

Biogas Investigation/Cogeneration

Twelve (12) samples of biogas were collected throughout the year and analyzed for siloxanes, methane, carbon dioxide, oxygen, nitrogen, hydrogen sulfide, and for BTU. Total Siloxanes as silocon averaged 17 ppbv ranging from 0 to 36. Hydrogen Sulfide concentrations averaged 3,385 ppmv ranging from 312 to 8,983. Energy values averaged 588 BTU/cubic feet (wet) ranging from 500 to 660.

Average annual values of analytical results from previous years are summarized as follows:

Year	Soloxanes (ppbv)	Hydrogen Sulfide (ppmv)	BTUs/cubic foot (wet)
2012	399	12,556	649
2013	136	7,331	639
2014	107	4,023	602
2015	NA	3,129	570
2016	159	3,266	559
2017	131	3,083	683
2018	72	4,043	591
2019	17	3,385	588

The new Stiles Facility Air Permit effective 9-16-2019 through 9-16-2024 was received on 10-2-2019. The Shelby County Health Department granted an extension of a construction permit through August 25, 2020 to allow the connection of a natural gas line to the unit. This will also involve re-commissioning the co-gen for service and then firing it with natural gas while projects are implemented to both increase biogas production and remove hydrogen sulfide.

Mississippi River Stage Impact

Approximately 3100 acres extending from North Second Street east to Hollywood Boulevard and from James Road south to Chelsea Street is subject to backwater conditions during elevated Mississippi River stages. River stage has major impact on Stiles Facility average daily flow beginning at 29 feet. A graph showing annual average daily flow versus number of days per year river stage exceeds 29 feet is provided in Figure 12. Total rainfall also impacts annual average daily flow. Using multivariate analysis, annual average daily flow = 0.23 x Days (> 29 feet) + 0.37 x Inches of Total Rainfall + 65.36 mgd from 1979 through 2019 at an R2 = 0.69. The annual frequency of occurrence is shown below:

Miss River Stage 1979 thru 2019		
Number of Days >29 feet	Annual Frequency	Interval
0	31.1%	3
1-10	4.7%	21
10-20	10.2%	10
20-30	23.4%	4
30-40	11.0%	9
40-50	7.1%	14
50-60	2.8%	36
60-70	5.6%	18
70+	4.1%	24
154	0.1%	1000

In 2019, there were 154 days when river stage was greater than 29 feet, a probability of occurrence of 1 in 1000 years. A typical year has 21 days above 29 feet river stage.

2017-2018 Flow and Inflow / Infiltration Study

Barge was asked to perform a flow study involving 41 flow meters collecting data from March 2017 through January 2018 in order to determine volume requirements for an equalization basin. The equalization basin would reduce rain-induced peak flow at the Stiles Facility to below 135 mgd. These data were reviewed to determine typical dry weather flow range for the 3 major interceptors coming into the Stiles Facility and then compared with typical influent flow range derived from the same time frame. This is possibly the first comprehensive look at Stiles Facility flow input that has been performed. The results are as follows including rate of rain-induced inflow / infiltration derived by Barge:

Interceptor (Monitoring Point)	ADDWF (range), mgd	Inflow / Infiltration rate, mg per inch rain
Wolf River Interceptor (WR01)	40-50	19.1
Mud Island Interceptor (FS01A)	6-8	3.13
KTG (C1O)	2-3	0.15
Loosahatchie Interceptor (LO1)	12-14	3.8
Lagoon Supernatant / Dewatering Filtrate Return / American Yeast (estimated)	3-4	1
Total Range	63-79	27.18

The dry weather range based on the flow monitoring study is 63 – 79 mgd with a median value of 71 mgd. The expected dry weather range during this same period based on treatment plant flow data is 55 - 80 mgd with a median value of 68 mgd. Based on the Barge Study, 63 percent of the dry weather flow comes from the Wolf River Interceptor. An overall inflow / filtration rate of 27 million gallons per inch of rain was calculated based on the Barge study. An overall 29.5 million gallons per inch of rain was calculated based on influent flow monitoring at the Stiles Facility. Based on the Barge Study, seventy percent of the inflow / infiltration received at the treatment plant comes from the Wolf River Interceptor.

Laboratory

The laboratory located at the M. C. Stiles Facility received samples from the Stiles Facility, the Maxson Facility, and Chapel Hill package treatment plant. The laboratory performed a total of 25,725 analyses in-house broken down as follows compared to past years:

Year	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
Stiles	12,953	13,019	10,089	13,157	11,775	10,322	16,280	NA	NA	10,117
Maxson	12,167	11,931	15,121	22,537	18,952	18,716	23,165	NA	NA	13,621
Chapel Hill	603	559	549	533	447	262	685	NA	NA	312
Industrial Monitoring	2	0	0	339	617	514	5303	NA	NA	2019
Total	25,725	25,509	25,759	36,566	31,795	29,814	45,433	41,946	51,993	26,069

Pretreatment Program

The program regulates the discharges of all industrial users, enforces the requirements of the sewer use ordinance, implements the federal categorical standards on specific process wastewaters, and manages the hauled waste program. In 2019, as part of regulating these industrial users, the Control Authority collected 295 samples, performed 105 site inspections, and issued 195 Notices of Violation. There were 99 significant industrial users (SIUs) that were monitored and reported on to the Approving Authority (TDEC). A summary of such statistics from previous years is as follows:

Year	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010

Samples	295	285	244	313	205	181	291	386	313	1714*
Inspections	105	107	98	105	93	101	93	149	93	75
NOVs	195	233	220	288	146	206	72	497	131	79
SIUs	99	104	101	98	98	95	97	101	94	95

*likely includes pH checks

For the year 2019, the City collected $365,222 for disposal of 2,416,765 gallons of hauled wastewater. Most of the hauled waste deliveries are septic and grease trap related. A summary of such statistics from previous years for comparison is as follows:

Year	Rate/gallon	Volume (gallons)	Revenue
2019	13 cents	2,416,765	$365,222
2018	13 cents	4,818,131	$626,357
2017	13 cents	4,649,995	$604,553
2016	13 cents	5,009,616	$651,160
2015	13 cents	8,550,823	$1,111,607
2014	13 cents	9,319,286	$1,211,507
2013	13 cents	6,007,191	$831,482
2012	13 cents	9,804,557	$1,139,951
2011	11 cents	14,025,493	$1,336,278
2010	8 cents	15,017,219	$1,046,637

The sewer fee history is summarized below:

Date	Volumetric	BOD Surcharge	TSS Surcharge	COD Surcharge	Ordinance
5-15-1979	28.52 cents	1.7 cents	0.9 cents	-	#2877
7-1-1981	-	3 cents	5 cents	-	#3127
1-1-1982	58.68 cents	2.7 cents	4.6 cents	-	-
7-1-2004	87.5 cents	3.65 cents	5.96 cents	-	#5036
7-1-2008	98.5 cents	3.86 cents	6.21 cents	-	#5239
7-1-2009	105.3 cents	-	-	-	#5312
8-1-2009	-	3.88 cents	6.61 cents	-	-
7-1-2010	226.7 cents	4.28 cents	7.16 cents	-	#5356
1-1-2018	287 cents	-	-	-	#5651
1-1-2020	332 cents	-	-	3.96 cents	#5651
1-1-2021	-	-	-	4.821 cents	-

The Maxson Facility rated capacity is 90 mgd for flow, 490,000 ppd for influent BOD and 410,000 ppd for influent TSS. The Stiles Facility rated capacity is 135 mgd for flow, 275,000 ppd for influent BOD and 374,000 ppd for influent TSS. Influent flow and loading trends are shown in Figures 13-15 for the Maxson Facility and in Figures 16-18 for the Stiles Facility.

For the Maxson Facility, Figure 13 shows annual average daily flow and average daily dry weather flow since 1976. Figure 14 shows annual average influent and industrial BOD loading since 1976 and 1989 respectively. Figure 15 annual average influent and industrial TSS loading since 1976 and 1989 respectively. For the Stiles Facility, Figure 16 shows annual average daily flow and average daily dry weather flow since 1978. Figure 17 shows annual average influent and industrial BOD loading since 1978 and 1992 respectively. Figure 18 annual average influent and industrial TSS loading since 1978 and 1992 respectively.

Since 1979, the Additional Treatment Cost charged to industries has increased at an annual rate of 1.72 % for BOD and 4.91% for TSS and the volumetric charge has increased at an annual rate of 5.8%.

3.0 Analysis and Future Outlook

Analysis

The Maxson Facility treatment capacity history is summarized as follows:

Year	Flow (mgd)	BOD (ppd)	SS (ppd)
1975	80	186,000	133,500
1985	80	390,000	335,000
1997	90 (not really)	490,000	410,000

The design for the Maxson Facility was based on sewer and industrial waste sampling conducted in 1967 as part of 201 Facilities planning. In 1972, with the passage of P.L. 92-500, nationwide 30-30 effluent discharge limits were established. The plant's design criteria pre-dated the 1972 legislation that only required 85% removal. An NPDES permit was issued effective 1-1-1975 and the plant began operations later that year. Records from November 1975 through June 1977 indicate influent monthly average loadings approaching 300,000 ppd for BOD and 250,000 ppd for TSS and monthly average removals of 80% for BOD and 62% for TSS.

On 7-22-1977, the U.S. DOJ/EPA filed suit against the City for violations of the discharge permit from 6-30-1975 through 7-22-1977. The Court entered the Consent Decree on 7-28-1978 to increase plant capacity. The Decree was modified on three subsequent occasions: Amended Decree (2-23-1979); Second Amendment to Decree (5-7-1979), and Consent Order (12-31-1981). The work was performed under ten contracts consisting of additional blower purchase (Blower #6), piping modifications, blower installation, electrical modifications, site preparation, three (3) primary clarifiers, administration building modifications, electrical reliability modifications, computer modifications, and secondary treatment modifications (4 ABF Towers). The Decree was terminated on 10-10-1986.

The Maxson Facility existing primary clarifier sweeps were modified during 1991-93 to increase sludge removal. A second capacity expansion took place 1995-1997 consisting of the addition of two ABF Towers, a 4th Primary Clarifier/Thickener, and two Secondary Clarifiers. A covered lagoon capacity expansion project was completed in 2004 which covered Lagoons 2A, 2B, and 2C and improved solids capture and primary sludge pumping. Primary sludge dewatering was discontinued in 2004 and the wasting of primary sludge directly to the lagoons began for co-digestion with waste activated sludge began. In 2006, the bio-towers began to be periodically operated in roughing mode for the first time as suggested by Dr. Orris Albertson with great success resulting in energy savings, improved effluent quality, and less waste activated sludge. This practice continued sporadically through mid to late 2019 and was discontinued due to valve problems. In 2010, Lagoon No. 1 was recovered and the influent head works replaced. In 2013, the Fine Bar Screens were put into operation and an immediate reduction in plastics and other undesirable items getting into primary and waste activated sludge was noted, a problem that had become more pronounced after the wasting of primary sludge to the lagoons began in 2004. The Primary Scum Removal System was reworked in 2014 increasing the overall removal efficiency of the primary clarifiers. Figure 19 shows BOD removal efficiency and Figure 20 shows TSS removal efficiency over the years. The Maxson Facility operated at consistent removal efficiency since 1987 after the original Consent Decree until about 2010. There has been a slight decline and more erratic performance since that time.

Co-digestion of primary sludge with waste activated sludge beginning in 2004 lead to a dramatic increase in biogas production that continued through 2016. In 2017, biogas production began falling off. There has been a decrease in activated sludge wasting in recent years due to increased use of roughing mode that may have contributed to the decline in biogas production. In addition, the buildup of inert solids in Lagoon 2A has likely been accelerating after the TVA ash release in 2010.

Based on population served (an estimated 404,570 people), the expected unmonitored influent BOD should be 69,000 ppd or 92,000 ppd based on minimum flow and 250 mg/l BOD. Figure 21 shows trends in unmonitored influent BOD. The increase in BOD returning to the head of the plant was totally unexpected after completion of major lagoon improvement projects which focused on solids retention. Similarly, the expected unmonitored influent TSS should be 81,000 ppd or 110,000 ppd based on minimum flow and 300 mg/l TSS. Figure 22 shows trends in unmonitored TSS. The monitored result includes the 24-inch dewatering filtrate / lagoon supernatant sampling point. Two other return flow lines including Lagoon No. 1 and Horn Lake Ditch are not monitored. The graph shows major spikes in unmonitored influent TSS occurring since 2004. The cause of these spikes is believed to be lagoon interference from the unmonitored discharge lines.

Average annual effluent BOD concentration since 1982 is shown in Figure 23. As shown below, effluent BOD concentration is typically a function of removal efficiency and influent loading. The influent loading data has been adjusted for BOD recycle from the lagoons beginning in 2004. The data shows the best and most stable effluent quality occurring at less than design loading when the removal efficiency is 97%.

		Effluent BOD mg/l		Influent BOD ppd	
Removal Efficiency	Number	Average	95 Percentile	Average	95 Percentile
93%-94%	6	30	40	251,619	388,911
95%	15	31	39	379,412	463,725
96%	14	27	35	363,896	502,500
97%	3	20	25	349,572	429,260

In 2019, adjusted for recycle, BOD removal efficiency = 93% with average effluent = 40 mg/l and loading = 399,000 ppd. This falls just at or slightly above the 95 Percentile range for concentration and loading for this removal efficiency.

Average annual effluent TSS concentration since 1982 is shown in Figure 24. As shown below, effluent TSS concentration is typically a function of removal efficiency and influent loading. The influent loading data been adjusted for TSS recycle from the lagoons beginning in 2004. The table shows that the Maxson Facility produces the best and most stable effluent quality when influent TSS is no greater than design loading and removal efficiency is 94% or greater.

		Effluent TSS mg/l		Influent TSS ppd	
Removal Efficiency	Number	Average	95 Percentile	Average	95 Percentile
90%-91%	4	39	48	250,833	355,197
92%	12	36	44	303,538	449,068
93%	7	29	39	256,950	379,308
94%	5	27	33	259,047	347,148
95%	7	30	35	370,070	497,905
96%	3	30	32	446,374	483,899

In 2019, TSS removal efficiency = 93% with average effluent = 42 mg/l and loading = 413,000 ppd. This is above the 95 Percentile range for this concentration and loading at 93% removal efficiency.

As shown in Figure 25, combined primary and secondary sludge wasting peaked in 2006. Beginning that year, as previously discussed, the bio-towers began to be periodically operated in roughing mode for the first time producing less waste activated sludge. The Maxson Facility has been operated mainly in ABF mode since mid to late 2019. As shown in Figure 26, annual power consumption peaked between 1997 and 2004. Power consumption began decreasing in 2005 as the plant began operating more in roughing mode resulting in greater efficiency of the bio-towers and less blower operation as well as improved effluent quality. This downward trend continued until 2010. Energy consumption has been increasing since then due to inability to maintain consistent roughing mode among other conditions that tend to increase blower usage.

The Stiles Facility has had no major process modifications since original construction other than discontinuing aerobic digesters in the early 1990s. Industries have come on line in recent years to bring the plant up to near full treatment capacity. In 2003, American Yeast began production in February and KTG began production in March. Discharges increased as these facilities ramped up to full production. In 2013, KTG expanded production with the startup of its TAD unit. Figure 27 shows BOD and Figure 28 shows TSS removal efficiency over the years. The figures show a possible decline in BOD removal efficiency after 2007 and in TSS removal efficiency after 1998.

Trends in unmonitored influent BOD loading are shown in Figure 29. The expected unmonitored influent BOD based on population this size (an estimated 427,435 people) is 73,000 ppd or 85,000 ppd based on minimum actual flow and 250 mg/l BOD. The lowering and increased stability of that number indicates that industrial discharges are being effectively monitored for that parameter. Unmonitored influent TSS has historically been erratic and higher than expected over the years but has been becoming steadier and trending upward as shown in Figure 30. The expected unmonitored influent TSS based on population this size (an estimated 427,435 people) is 85,000 ppd or 103,000 ppd based on minimum actual flow and 300 mg/l TSS.

Average annual effluent BOD concentration since 1982 is shown in Figure 31. As shown below, effluent BOD concentration is typically a function of removal efficiency and influent loading.

The table shows that the Stiles Facility produces the best and most stable effluent quality when its removal efficiency is 93-96% and influent loading slightly lower than design.

		Effluent BOD mg/l		Influent BOD ppd	
Percent Removal	Number of Years	Average	95 Percentile	Average	95 Percentile
79-88%	4	42	56	247,548	263,503
89-90%	6	30	37	230,440	262,611
91%	6	32	38	245,703	270,376
92%	8	26	32	224,868	266,376
93%	9	21	25	223,590	240,064
94%	5	20	23	237,052	250,960

In 2019, BOD removal efficiency = 81% with average effluent = 59 mg/l and loading = 265,000 ppd which falls just outside the 95 Percentile range of historic performance for concentration and loading for 79%-88% removal efficiency. Adjusted for PAA impact, the BOD removal efficiency = 83% with average effluent = 53 mg/l which is within the 95 Percentile range for concentration at that removal efficiency.

Average annual effluent TSS concentration since 1982 is shown in Figure 32. As shown below, effluent TSS concentration is typically a function of removal efficiency and influent loading. The table shows that the Stiles Facility produces the best and most stable effluent quality when its removal efficiency is 94-96% and the influent loading is considerably below the design loading.

		Effluent TSS mg/l		Influent TSS ppd	
Percent Removal	Number	Average	95 Percentile	Average	95 Percentile
85%-89%	5	40	46	284,582	312,360
92%	7	28	38	263,547	317,878
93%	5	28	33	269,619	301,806
94%	14	21	28	265,523	308,714
95-96%	7	17	20	262,834	304,903

In 2019, TSS removal efficiency = 87% with average effluent = 40 mg/l and loading = 296,000 ppd which falls within the 95 Percentile range of historic performance for concentration and loading for 85%-89% removal efficiency.

As shown in Figure 33, sludge wasting peaked in 1988 and was subsequently on the decline for many years reaching a low point in 2003 when both American Yeast and KTG both began production. Sludge production then picked up mirroring the discharges of these two industries. Presumably, American Yeast's high strength organics was converted to solids and wasted from the activated sludge process and KTG's mainly very low volatile solids were direct wasted to the lagoons. After the economic slowdown, sludge wasting began picking back up in 2014, helped by the increase influent solids loading after the addition of the TAD unit at KTG. As shown in Figure 34, energy consumption was on a 20 year decline starting in 1981 as the plant implemented the Aeropt system and discontinued aerobic digesters. Energy consumption began increasing in 2003 when both American Yeast and KTG began production. Aeropt was discontinued in 2010 and energy consumption remained flat for several years. Energy consumption has increased in recent years due to increased blower usage in an attempt to improve effluent quality, especially upstream of PAA addition in more recent years.

The cost that MLGW charges for electricity has been in the news lately with talk about MLGW severing ties with TVA. Essentially, the wastewater treatment plants pay a wholesale rate from TVA. The average annual all-in cost for main power going back to 1996 is shown in Figure 35. Approximately 38% of the cost is estimated to be for generation based on utility bills. This would put the 2019 wholesale rate for power generation paid by the City's wastewater treatment plants at 2.62 cents per kwh. The MISO average wholesale rate for power generation in 2019 was around 3.1 cents. Based on this analysis, the City would be better off staying with MLGW and increasing participation in curtailment programs.

Table 1 and Table 2 both show influent concentration and loading and effluent concentration for BOD, COD, and TSS over the years. Table 3 compares total influent flow, average daily dry weather flow, and average daily billed wastewater flow at both treatment plants compared with drinking water production for each month. The total influent flow is the average daily flow received at the treatment plants. The average daily dry weather flow is based on days with no apparent river or rain impact. The average daily billed wastewater flow is based on residential and commercial water usage and billed large industrial user and wholesale customer sewer usage. Drinking water production is the metered output of the water treatment plants that is sold to residential and commercial customers and a small portion of it is not affiliated with the City. Some industrial users use private wells to generate cooling and process water instead of using MLGW drinking water which is discharged to the sanitary sewer on a continuous basis and may constitute 10% or more of dry weather flow received. Figure 36 shows the Table 3 data over the last 10 years.

Tables 4 and 5 present average annual rain-induced flow per inch of rain, percent of average daily flow that is rain-induced, average daily flow, average daily dry weather flow, annual rainfall, and percent of average daily flow that is river-induced for each treatment facility since 1975 and 1979 for the Maxson and Stiles Facilities, respectively. Trends in rates of rain-induced flow are shown in Figures 37 and 38. Over the last 10 years, rain-induced flow has averaged 27 million gallons per inch of rain at the Maxson Facility and 27 million gallons per inch of rain at the Stiles Facility. Trends in average daily and average daily dry weather flow for both facilities combined are shown in Figure 39. Overall, rain-induced flow has comprised 5.7% and 4.7% of average daily flow received at Maxson and Stiles, respectively, and river-induced flow has comprised 3.8% and 13.5% of average daily flow received, respectively. Combined ADF peaked in 1998 and Combined ADDWF peaked in 1996. Both have been on a steady decline with the exception of the last couple of years.

Data related to plant operations and sludge production is presented in Tables 6 and 7. Some additional tables presented are Table 8, the monthly plant operating budget expenditures and Table 9, a breakdown of plant expenditures by category. Treatment costs averaged 68 cents per

1000 gallons in 2019, an increase of 14 cents from the previous year as shown in Figure 40 with previous years for comparison. Tables 10 and 11 comprise the largest industrial user's monthly fees for each plant. Table 12 shows the three wholesale customer monthly fees. These tables show that the fees from these 19 industrial users, hauled wastes, and 3 wholesale customers cover all of the operating cost of the plants, thus allowing the City to maintain low stable rates. Of the $46,959,040 in operating expenditures in 2019, these industrial, hauled waste, and wholesale customers' fees of $38,142,230 amounted to 81.2% of the costs. The total fees received decreased $2,080,405 for the 19 industrial users and hauled waste customers and increased $79,433 for the three wholesale customers from the previous year. Table 13 shows annual hauled waste volume and revenue over the years.

Table 14 is a monthly comparison of the utility charges for both treatment plants. The treatment plants changed to Time of Use contracts effective 10-1-2015 which makes unit cost tracking more complicated due to different on-peak and off-peak time blocks.

Table 15 is a mass balance of the origin of the total suspended solids in the Maxson Facility influent. This table shows that the lagoon supernatant was 4% of nominal plant TSS loading in 2019. Table 16 shows the relationship between total suspended solids in the lagoon supernatant to the pounds of sludge wasted to the lagoon over time. Table 17 shows the total loading of BOD_5 of the plant and the amount contributed by the major industries. Table 18 shows the total loading of COD of the plant and the amount contributed by major industries. Table 19 shows the total loading of total suspended solids of the plant and the amount contributed by the major industries. Table 20 shows the water usage and cost. Table 21 is a table showing the electrical usage, cost, biogas production, and propane usage for the covered lagoon. Table 22 is a table showing the electrical usage and cost for other electrical meters. The City uses an average of 1.7 mgd of potable water at the Maxson Facility for cooling equipment and dewatering spray down and could be asked in the future to transition to a lower quality water source based on future industrial need and potential new regulatory requirements. Table 23 shows long-term trends in BOD loading and wasting.

Table 24 shows the utility usage, cost, and biogas production for the Stiles Facility. Table 25 shows the plant water usage and cost as well as the electrical usage and cost for dewatering. Table 26 shows monthly electrical usage and costs at Chapel Hill. Table 27 shows effluent concentration and compliance history for Chapel Hill. Table 28 shows total loading of BOD_5 for the Stiles Facility and the amount contributed to it by the major industries. Table 29 shows the total loading of COD of the plant and the amount contributed by major industries. Table 30 shows the total loading of TSS for the plant and the amount contributed by the major industries. The City uses an average of 0.68 mgd of potable water at the Stiles Facility for cooling equipment and dewatering spray down and could be asked in the future to transition to a lower quality water source based on future industrial need and potential new regulatory requirements. Biogas purchases by American Yeast are shown in Table 31.

Table 32 summarizes the electrical usage, total cost, unit charges, and savings over the past several years for both treatment facilities. Table 33 and Table 34 show comparisons of the Stiles Facility and Maxson Facility influent flows versus the combined flows of the major industries.

Table 35 shows requisitions and purchase orders for goods and services by both plants during 2019 totaling $1,071,979. Other goods and services were purchased using negotiated contracts which are not captured in the requisition process.

Regulatory Developments

Final NPDES permits for both treatment plants were issued on 12-2-2011 for the Stiles Facility and on 12-6-2011 for the Maxson Facility, effective 1-1-2012 and expiring on 12-31-2016. Certain elements of the permits were subsequently appealed. TDEC proposed a modification to the Maxson Facility permit in September 2012 seeking to establish a compliance schedule for disinfection keyed to a revised effective permit date. However, the proposed modified permit was never issued. Applications for new permits were submitted 7-5-2016. The City met with TDEC on 4-10-2017 to review the issuance of new permits. TDEC presented information at the meeting indicating the requested secondary treatment adjustments (i.e. variances) would not be granted. On 10-20-2017, the City submitted a report explaining why the variances should be granted as a supplement to the previously submitted permit applications. New draft permits for both treatment plants were public noticed on 11-13-2018. The permits stuck to the nationwide 30-30 standards and did not allow Best Professional Judgement (BPJ) to be used for industry groups without promulgated categorical effluent limits for secondary treatment adjustment. Public Works officials drove to Nashville on 12-10-2018 to meet with TDEC and learned that TDEC was not be flexible on this issue. The City was granted an extension request for public comment to 2-15-2019. A public meeting for the Maxson and Stiles Facility proposed NPDES permits was held on 2-26-2019. TDEC would not impose a permit limit on PAA residual since the City would have an incentive to keep it low due to cost. Any permits issued by TDEC of course would need to be approved by US EPA. The City submitted comments on 3-20-2019, totaling 800+ pages. On 4-18-2019, Vojin Janjic emailed that based on the comments received both NPDES permits would be withdrawn and re-issued at a later date which would require a new public notice. On 7-15-2019, TDEC had a public hearing on proposed amendments to Chapter 0400-40-05 to better reflect the Department's implementation of federal NPDES permitting requirements. The City's attorney Gary Cohen responded with comments on 7-25-2019. It is believed that additional rulemaking will be introduced in the upcoming months in reference to secondary treatment adjustments with the City of Memphis in mind. Therefore, re-issuance of the Maxson and Stiles Facility NPDES permits will be delayed until such rulemaking is passed.

Two permitting issues loom on the horizon for both treatment plants. First, a new method demonstrating compliance with 40 CFR 133.102 as it relates to treatment and performance will need to be negotiated with TDEC and agreed upon by the US EPA. At least for the Stiles Facility, the permitting issue will be how to account for the BOD contribution of PAA to final effluent using the existing regulatory compliance point "final effluent" sample location (downstream of the disinfection tank.) For the Maxson Facility, more than likely, secondary treatment will be based on the existing final effluent sample collected upstream of the disinfection tank and PAA impact on BOD would not be considered.

Second, there is the issue of not qualifying for a secondary treatment standard adjustment for effluent limits. Federal secondary treatment standards allow for an effluent BOD or TSS credit in mg/l to be added to the 30-30 limit for industrial categories that exceed 10% of design flow or loading of the POTW. The City has lost large industrial users over the years with categorical treatment standards with numerical limits. These have been replaced with large industrial users in categories without numerical limits. TDEC has been inconsistent in the past about allowing the use of industrial categories with BPJ-derived numerical limits, as discussed below.

Past permit limits are shown in Table 36. The original permits for both treatment plants were issued 11-30-1974 based on 30-30 monthly and 45-45 weekly averages for BOD-TSS. The City began negotiating with TDEC in 1979 for a variance from the 30-30 limits based on its large industrial wastewater contribution as part of it permit re-application. In a US EPA memo dated 3-11-1980 from Frank Hall to Sanford Harvey of Region 4, the US EPA stated that variances would not be granted for industrial categories without numerical limits based on the premise they were not public noticed. TDEC subsequently worked with the City by issuing permit limits in 1982, 1984, and 1987 allowing the use of Best Professional Judgement (BPJ) for industrial categories without promulgated effluent guidelines because they believed the permit rationale would satisfy the public notice requirement.

Beginning in 1993, TDEC took the opposite stance and disallowed the BPJ approach, stating that the previous permits were issued in error. There was another twist. TDEC included a review of past performance in the permit rationale, which had not been a consideration in previous permits but now played a role as part of anti-backsliding. In 2000, permit limits were issued in the same manner. The City did not challenge the rejection of BPJ apparently because there were categorical industries with numerical limits with sufficient volume of discharge to qualify for reduced limits.

In 2011, in the case of the Maxson Facility, TDEC <u>allowed</u> the substitution of Solae in the Grain Mills Industry Category (40 CFR 406) using the BPJ approach. This boosted the secondary treatment adjustment above the previous permit limits and allowed the previous permit limits to continue by default. If this approach had not been taken, the Maxson Facility would have been bound by the 30-30 standards and be in non-compliance upon issuance of these new limits. The Stiles Facility permit limits, on the other hand, did not need to involve BPJ in the secondary treatment adjustment calculations since sufficient qualifying industrial categories with numerical limits were available and were issued only slightly lower than previous permit limits. While the Stiles Facility had a slight reduction in permit concentration limits, it saw the elimination of daily, weekly, and monthly "pounds" as permit limits. This resulted in fewer opportunities to violate compared to the Maxson Facility.

A third Maxson Facility expansion project was initiated in 2104 at the encouragement of City officials to facilitate expansion of existing industries as well as establishment of new industrial development in the Pidgeon Industrial Park. During the early years of the City of Memphis Pretreatment Program, discharge agreements were signed with industrial users and industries sought to "negotiate" greater discharge limits than currently needed for future expansion. In later years, the City moved to a permit approach which had less perceived opportunities for "negotiation" and began to reign in past promises to industries in an informal effort to avoid over allocation compared to rated treatment capacity. As such, industries began contacting the Mayor's office to secure such commitments outside of the industrial user permit system. For the moment, an informal truce has been reached between existing industries and the wastewater treatment plants with both treatment plants near capacity and able to meet current permit limits during normal operating conditions. An expansion would allow existing industries to be able to plan for future expansion and allow the City to recruit new industries to the Pidgeon Industrial Park where there is plenty of land and other resources available.

The quest for expansion began as an $85 million Guaranteed Maximum Price (GMP) proposal from CDM Smith. Over time, the expansion project was broken into five smaller separate GMPs which, taken to together, would encompass the desired capacity increase and performance improvements as well as disinfection and influent coarse bar screen replacement. The cost to perform all of this work will be approximately $251 million once all of the construction packages have been developed with approximately $83 million funding from SRF. The upgrades will go a long way toward ensuring compliance in a worst case regulatory scenario (i.e. 30-30 limits). Any capacity increase associated with the upgrades could potentially cause the City to lose its eligibility for secondary treatment adjustment whether or not BPJ is allowed because the industrial discharge likely would be below the 10% rated capacity threshold for any one industrial category.

Such a capacity increase and/or performance improvement would need to be implemented at the Stiles Facility for the same reasons (i.e. 30-30 limits) due to the potential to loose eligibility for secondary treatment adjustment. This would be the result of the reduction of flow and loading from existing industries with categorical limits and the current rejection by TDEC of the BPJ approach for industries without promulgated categorical effluent numerical limits. It is a strong possibility that this problem will "take care of itself" once the COD surcharge is implemented on 1-1-2020 and industries begin cutting back their discharge as a result. Other creative measures will need to be implemented such as co-digestion of American Yeast beer waste in the Stiles Facility covered lagoons. Otherwise, primary clarifiers would need to be added to reduce the loading going to secondary treatment.

On 11-27-2007, the Natural Resources Defense Council petitioned the US EPA to request updated information about secondary treatment nutrient removal capability and establish new technology-based nutrient limits as part of the secondary treatment standards. On 3-16-2011, a Nutrient Reduction Framework (NRF) memo was issued to the EPA regional offices to guide EPA technical assistance and collaboration with states. This initiative was subsequently rolled down to the states. On 12-14-2012, the EPA responded to the NRDC petition saying that it agreed that insufficient data exist to draw any general conclusions about the ability of secondary treatment to remove nutrients and that a uniform set of nationally applicable, technology-based nutrient limits was not warranted at this time. In February 2016, the EPA issued the 1st Gulf of Mexico Hypoxia Task Force report. On 10-16-2019, EPA issued the 2nd Gulf of Mexicon Hypoxia Task Force report. Both the Maxson and Stiles Facilities were listed as major point source contributors from Tennessee.

As a result, the EPA initiated a multi-year, multi-phased national study in nutrient discharges from POTWs in order to 1) establish a statistically representative, nationwide baseline for nutrient discharge and removal at these facilities, and 2) characterize operational and management practices that result in improved nutrient reduction. A questionnaire was published in the Federal Register on 9-19-2016 but was withdrawn apparently after the presidential election. The City responded to a POTW Nutrient Survey / Voluntary Questionnaire (online) issued by the EPA on 10-22-2019 for both treatment facilities.

In 2014, TDEC submitted a draft NRF to the EPA for review and subsequently began implementing it. The TDEC NRF prioritizes HUC10 watersheds with 303(d) listed segments. Tennessee is also developing a Non-Point Source nutrient reduction strategy for Mississippi

River basin (Hydrogeological Unit IV) using a farmer-led approach so it is not known at this time how the Maxson and Stiles Facilities will be affected. The initial state-wide implementation of nutrient reduction will be an emphasis on optimization instead of capital improvement. In 2018, TDEC's emphasis was on developing a larger stake holder group for input on the NRF in general and developing site-specific criteria for three reservoirs in Tennessee as a priority. As of 2019, sub-working groups are being formed and additional stakeholders are being sought at the direction of the governor before new rulemaking can be proposed.

PAA

Continuous PAA dosing began on 11-10-2018 at the Stiles Facility and continued through the year at cost of $13,762,445.66 or $37,705 per day. This equates to 34.9 cents per thousand gallons treated at an average dose of 13 mg/l. The average pre-final effluent color was 726 Pt Co.

The average effluent color for the Maxson Facility measured in 2019 was 408 Pt Co. Based on this effluent color concentration, an average PAA dose of 7.3 mg/l is expected. This equates to 19.6 cents per thousand gallons treated. Based on a long-term ADF of 70 mgd, the average annual cost for PAA at the Maxson Facility is estimated to be $5 million per year.

Metals

The Pretreatment Program had set allowable limits for many categorical dischargers that resulted in the significant reduction of the amount of metals received over the past three decades. With the promulgation of the 503 Sludge Regulations, the amount of various metals entering the plants became a greater regulatory concern. Prior to 1994, the discharge of metals had been controlled and regulated but there were no regulations that defined allowable concentrations of various metals in biosolids. Table 37 shows the metals concentrations over the years in the plants' influent and biosolids. The tables utilize the combination of the six most common metals, zinc, copper, nickel, chromium, lead, and cadmium. As the table shows, influent metals loadings have generally decreased over the past 4 decades.

Table 37 shows the average limit that would apply to the Maxson and Stiles Facilities either as land application or surface disposal under the 503 regulations as well as the metals concentrations for biosolids sampled in 2019. The metals are arsenic, cadmium, chromium, copper, lead, mercury, molybdenum, nickel, selenium, and zinc. Metal concentrations in surface-disposed bio-solids have decreased over the past 3 decades at both treatment plants along with influent metals loadings. In recent years, however, metals concentrations in surface disposed biosolids appear to be more of a function of depth and location of lagoon dredging than contemporaneous influent metals loading.

Personnel

There appears to be reluctance on the part of TDEC to give its blessing to any training or work-study program related to water/wastewater and have a central role other than giving credit for traditional correspondence courses. Being a right to work state, Tennessee has no regulations for apprenticeship programs like most northern and some western states. Instead, Tennessee supports "workforce development programs" that favor state-funded educational institutions. This has resulted in a hodge-podge of certificates awarded to students rather than degrees or specific licenses. Tennessee does have free 2-year community college and Pellisssippi State

Community College introduced a Water Quality Technology Program for Water/Wastewater in July 2019 with a wastewater treatment operator curriculum bestowing an Associate Degree in Technology after the completion of 60 semester hours. The student is still required to apply and take the state wastewater exam in accordance with state requirements.

Public Works has been unsuccessful using on the job training for an employment system as exemplified by the 369D MOU with its apprenticeship-like step raises. Since neither the City nor Tennessee has given legal standing to apprenticeship programs, the City has the option to create a non-union trainee position with a 36-month probationary status that would be the entry point to the wastewater treatment operator position. The recruitment of maintenance mechanics is equally problematic and will likely require the development of a similar program.

During 2019, one engineer obtained his Grade 4 license and one trainee obtained his Grade 3. As of the end of 2019, none of the 7 trainees passed the Grade 4 license exam. An eighth trainee entered the program late in the year already possessing a Grade 4 license and so will exit the program in 2020 after a 6 month minimum probation.

Employees of the Month are shown in Table 38. The employee of the month program was discontinued in 2016 in favor of giving out awards and spot bonuses. Table 39 shows the yearly number of budgeted positons for each plant over the years.